

CSP650-PROJECT
Software Requirements Specification
For Centralized Web-Based System for Stress Induction Methods

SEMESTER MARCH – AUGUST 2026

GROUP: M3CS2666A

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Ahmad Emir Erfan bin Mohd Zubri	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

1.0 Document Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Centralized Web-Based System for Stress Induction Methods.

The purpose of this document is to provide a detailed description of the system's functionality, performance requirements, and operational constraints. The system is designed to assist researchers in conducting stress-induction experiments by providing a centralized platform for managing participants, administering stress-induction methods, recording responses, and generating experiment data.

This document serves as a reference for developers, researchers, project supervisors, testers, and other stakeholders involved in the development and implementation of the system. It ensures that all parties have a clear understanding of the system's objectives, features, and requirements.
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1.1 Product Scope

The Centralized Web-Based System for Stress Induction Methods is a web-based platform developed to support researchers in conducting stress-induction experiments in a structured and efficient manner.

The system centralizes different stress-induction methods into a single platform where researchers can manage participant information, conduct experimental sessions, and collect research data. By digitizing these processes, the system helps reduce manual errors, improve data organization, and enhance the efficiency of research activities.

The platform allows researchers to create participant profiles, initiate stress-induction tests, monitor experiment progress, and store experiment results in a centralized database. This centralized approach ensures better accessibility, consistency, and management of research data.

1.2 Intended Audience and Document Overview

This SRS document is intended for stakeholders who are involved in the development and use of the Centralized Web-Based System for Stress Induction Methods.

- **Developers**: Developers will use this document as a guideline for designing, implementing, and maintaining the system according to the defined requirements.
- **Researchers**: Researchers will use the system to manage participants, conduct stress-induction experiments, and collect research data.
- **Project Supervisors**: Project supervisors will review this document to ensure that the system meets the research objectives and project requirements.
- **Testers**: Testers will use this document to verify whether the system meets the specified functional and non-functional requirements.
- **Participants**: Participants are individuals involved in the stress-induction experiments who interact with the system during experimental sessions.

1.3 Definitions, Acronyms and Abbreviations

Terms	Definition
SRS	Software Requirements Specification.
Researcher	A user responsible for conducting experiments and managing research data
Participant	An individual who participates in the stress-induction experiment
Stress Induction Method	A controlled experimental technique used to induce stress for research purposes
Experiment Session	A session where participants complete stress-induction tasks
System	Centralized Web-Based System for Stress Induction Methods.

1.4 Document Conventions

This document follows the IEEE Software Requirements Specification (SRS) standard format. The document is organized into sections that describe the system overview, functional requirements, non-functional requirements, and system features.

Consistent formatting and terminology are used throughout the document to ensure clarity and readability for all stakeholders involved in the project.

1.5 References

The development of the Centralized Web-Based System for Stress Induction Methods is based on the requirements of the Final Year Project and research materials related to stress-induction methodologies.

2 Overall Description

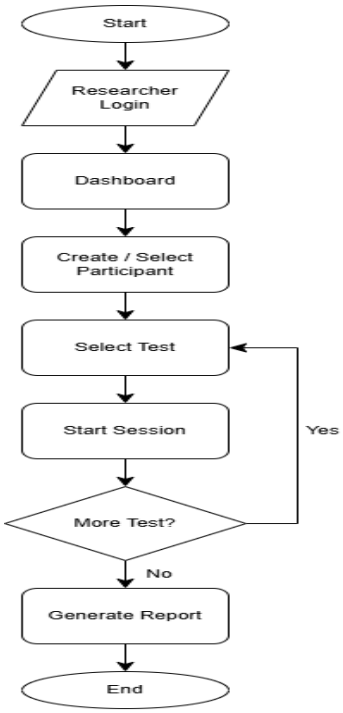
2.0 Product Perspective

The Centralized Web-Based System for Stress Induction Methods is a centralized system developed to assist researchers in conducting stress-induction experiments efficiently and effectively. The system aims to improve the current methods of conducting the test which the current method uses manual stress test and manual recording.

By centralizing the stress-induction methods in one single platform, the system allows researchers to manage the participant, conduct different type of test in one platform which can reduce the risk of data loss, inconsistency, and human error.

They system consists of the following major components:

- User Management Module: Handles authentication and role-based access control for system users, including administrators, and researchers.
- Participant Management Module: Stores participant information such as participant ID, participant information, and session records.
- Experiment Session Module: Guide participants through the experiment process by presenting stimuli, recording responses, measuring reaction times, and monitoring sessions progress.
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- Data Storage and Processing Module: Stores experiment data securely and performs automatic calculations such as reaction time averages, error rates, and stress scores.
- Reporting and Export Module: Generates experiment information and allows researchers to export raw data and reports in formats such as CSV for analysis and PDF for summary report.



2.1 Product Functionality

The Centralized Web-Based System for Stress Induction Methods offers the following major functions:

1. User Management:

    - Allow researchers and administrators to securely log into the system.
    - Manage user roles such as administrator and researcher.
    - Control access permissions to ensure only authorized users can configure experiments or export data.

2. Participant Management:

    - Create and store participant profiles using pseudonymous participant IDs.

3. Stress Test Management:

    - Guide researchers through the experiment process step-by-step.
    - Present stress-induction stimuli to participants during the session.
    - Monitor experiment progress in real time.
    - Allow participants to repeat experiment sessions while storing each session separately.

4. Reporting:

    - Generate automatic experiment summary reports.
    - Provide raw data.
    - Allow export in formats such as CSV and PDF.

2.2 Users and Characteristics

The Centralized Web-Based System for Stress Induction Methods is designed to be functioned to multiple users with different roles and responsibilities. The characteristics of these users are detailed below:

1. Administrator

    - Characteristics: Responsible for managing system configuration, user accounts, and system settings. The administrator ensures that the platform operates securely and efficiently.
    - Frequency of Use: Moderate
    - Privilege Levels: Full administrative access, including managing researcher accounts, configuring system settings, monitoring system activity, and maintaining data security.
    - Technical Expertise: Advanced knowledge of system administration and database management.

2. Inspector

    - Characteristics: Responsible for conducting stress-induction experiments using the system. Researchers configure experiment templates, manage participant information, initiate experiment sessions, and analyze experiment data.
    - Frequency of Use: High

- Privilege Levels: Authorized to create and manage experiment templates, manage participant records, conduct experiment sessions, view experiment results, and export research data.
- Technical Expertise: Moderate, including basic system navigation, experiment configuration, and data analysis tasks.

3. Participant

- Characteristics: Individuals who participate in stress-induction experiments conducted by researchers. Participants interact with the system only during experiment sessions when responding to stimuli presented by the system.
- Frequency of Use: Low to moderate
- Privilege Levels: Limited access to interact with experiment tasks during testing sessions. Participants cannot access system management features.
- Technical Expertise: Basic computer usage skills required to interact with experiment tasks and respond to stimuli.

2.3 Operating Environment

The Centralized Web-Based System for Stress Induction Methods operates within a secure and centralized environment. The following describes the hardware, software, and subsystems that support the system:

1. Hardware Platform
    - Server Requirements: The system will run on a dedicated server with a minimum of:
        - Processor: AMD Ryzen 5 or equivalent
        - RAM: 12 GB
        - Storage: 500 GB SSD
    - Client Devices:
        - Desktop computers or laptops for administers and researchers
        - Desktop computers or laptops for test sessions with participants

2. Operating System and Versions
    - Server OS: Windows Server
    - Client OS: Windows 10/11 for desktops and laptops

3. Operating System and Versions
    - Database Management System: MariaDB
    - Web Server: Apache HTTP Server
    - Application Framework: PHP
    - User Interface: Browser-based interface optimized for Chrome, Firefox, and Edge

2.4 Design and Implementation Constraints

The development of the Centralized Web-Based System for Stress Induction Methods must address several constraints that will limit the choices available to developers. These constraints are categorized below:

Type Of Constraints	**Constraints**
Hardware	Test sessions must be conducted using desktop computers or laptops to ensure accurate display of stimuli and reliable measurement of participant responses and reaction times.
Network	The system requires a stable internet connection to ensure experiment data can be recorded and stored without interruption. Network instability may affect the experiment session.
Software	The system must be developed as a web-based application using PHP, MariaDB database, and accessed through modern web browsers.
Security and Privacy	Participant data must be stored securely, and access must be restricted to authorized researchers and administrators only.
Language Requirements	The application must be developed using the English language to ensure consistency and usability for researchers

2.5 User Documentation

The Centralized Web-Based System for Stress Induction Methods will include comprehensive user documentation to support all user groups in effectively operating the system. The documentation components are as follows:

1. User Manuals
    - Administrator Manual: Includes instructions for managing user accounts, maintaining system settings, and monitoring system activities.
    - Researcher Manual: Provides detailed guidance on managing participant information, configuring experiment templates, conducting stress-induction sessions, and generating experiment reports.

2. Tutorials
    - Interactive Tutorials: Step-by-step walkthroughs that guide researchers on how to create participants, configure experiments, and run experiment sessions.
    - Video Tutorials: Visual demonstrations of key system features, such as starting experiment sessions, recording participant responses, and exporting experiment data.

3. Quick Reference Guides
    - One-page reference documents summarizing essential tasks for system users, such as creating experiment templates, running experiment sessions, and exporting research data.

2.6 Assumptions and Dependencies

The development and successful operation of the Centralized Web-Based System for Stress Induction Methods are based on the following assumptions and dependencies. Any changes to these factors may significantly affect the project's development and system performance.

Assumptions

1. Hardware Availability:

    - All end-user devices (desktop computers or laptops) meet the minimum hardware requirements such as at least 8GB RAM and a modern processor to ensure smooth system performance during experiment sessions.

2. Network Infrastructure:

    - The operating environment will have a stable and reliable internet connection to ensure that experiment data can be transmitted and stored in the database without interruption.

3. User Access and Training:

    - Researchers and administrators will receive sufficient guidance or training to operate the system effectively.
    - All system users will have unique login credentials and are responsible for maintaining the security of their accounts.

4. Software Compatibility:

    - The system will operate on supported operating systems such as Windows 10/11 or macOS.
    - Modern web browsers such as Brave, Google Chrome, Firefox, and Microsoft Edge will remain compatible with the system throughout its usage.

5. Organizational Support:

    - The research team and project supervisor will provide the necessary support and feedback to ensure the successful development and testing of the system.

6. Research Environment:

    - Experiment sessions will be conducted using desktop computers or laptops to ensure accurate presentation of stimuli and proper measurement of reaction time.
    - Participants will follow the instructions provided by the researcher during experiment sessions.

Dependencies

1. Database Management System (DBMS):

    - The system depends on the proper configuration and availability of the MariaDB database management system for storing participant data and experiment results.

2. Web Server and Hosting Environment:

    - The system requires a web server and hosting environment capable of supporting the web application and maintaining reliable access for researchers.

3. Development Tools:

    - The development of the system depends on tools such as Visual Studio Code for coding and Draw.io for system design diagrams.

4. Browser Compatibility:

    - The system depends on modern web browsers to function properly, including Brave, Google Chrome, Firefox, and Microsoft Edge.

5. Data Security and Privacy:

    - The system must follow appropriate data protection practices to ensure that participant information and experiment results are stored securely and accessed only by authorized users.

6. Stakeholder Feedback:

    - Continuous feedback from stakeholders, including researchers and the project supervisor, is necessary to ensure the system meets research requirements and project objectives.

3 System Features

3.1 USE CASE DIAGRAM



3.2 USE CASE DESCRIPTION

3.2.1 UC001 Create Account

Use Case ID:	UC001		
Use Case Name:	Create Account		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Inspector, Supervisor, Maintenance staff, Manager	
Description:	This use case describes the process of creating a new account in the Key Management System. Authorized actors can add a new user by providing essential details, such as username, email, role, and password.	
Preconditions:	The actor is logged into the system with valid credentials. The actor has the appropriate permissions to create accounts.	
Postconditions:	A new account is successfully created and stored in the system.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. The actor navigates to the "Create Account" page. 2. The actor inputs the required details, including: • User name • Email address • Role (Inspector, Supervisor, etc.) • Temporary password	1.1.The system validates the input details. 2.1.The system checks for duplicate usernames or emails. 2.2.If valid, the system creates the new account.

	3. The actor submits the form.	3.1.The system sends a notification to the new user with account details. 3.2.The system displays a success message to the actor.
Alternative Courses:	1. The system detects missing required fields and displays an error message prompting the actor to complete them. 2. The system detects an existing account with the same username or email and displays an error message: "Account already exists. Please choose a different username or email." 3. If the actor provides invalid data (e.g., email in an incorrect format), the system displays an appropriate error message and highlights the fields that need correction.	
Exceptions:	2.2 If the system encounters a technical error (e.g., database connection failure), it displays a message: "Unable to create account at this time. Please try again later."	
Includes:	Validate User Input (a common sub-use case for validating form data).	
Notes and Issues:	Ensure that passwords meet security standards (e.g., minimum length, complexity). Role selection must automatically assign the correct permissions for the new account. Consider implementing multi-factor authentication for added security.	

3.2.2 UC002 View Account

Use Case ID:	UC002		
Use Case Name:	View Account		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Inspector, Maintenance Staff, Supervisor, Manager	
Description:	The account information includes personal details, account activity, and assigned roles or permissions.	
Preconditions:	The actor must be logged into the system with valid credentials. The actor must have appropriate permissions to view account information.	
Postconditions:	The requested account details are displayed on the screen.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Logs into the system.	1. Validates the login credentials.
	2. Navigates to the "View Account" section.	2. Display account information.
Alternative Courses:	If the actor cancels the operation, the system returns to the main menu or previous page.	
Exceptions:	If the system fails to retrieve the account information due to a database issue, it displays a failure message and logs the error.	
Includes:	Search functionality for filtering accounts by name, ID, or other parameters.	
Notes and Issues:	Ensure that sensitive information (e.g., passwords) is not visible to unauthorized actors.	

3.2.3 UC003 Update Account

Use Case ID:	UC003		
Use Case Name:	Update Account		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The Supervisor updates the account information for an existing user in the system. This may include editing details such as name, contact information, or access privileges.	
Preconditions:	The user account must already exist in the system. The Supervisor must be logged into the system with sufficient privileges.	
Postconditions:	The account information is updated successfully in the system. A confirmation message is displayed to the Supervisor.	
Priority:	High	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Logs into the system.	1. Authenticates the Supervisor's credentials.
	2. Navigates to the "Account Management" section.	2. Displays a list of existing accounts.
	3. Selects the account to update.	3. Displays the account details.
	4. Edits the required fields and submits the changes.	4. Validates the updated information and saves the changes to the database.
	5. Receives a confirmation message of a successful update.	5. Logs the update activity for auditing purposes.

Alternative Courses:	If the Supervisor enters invalid data during the update, the system displays an error message and prompts the user to correct it.
Exceptions:	The system fails to update due to a database issue, in which case an error message is displayed, and the Supervisor is advised to try again later.
Includes:	Validation of updated fields
Notes and Issues:	Ensure proper logging of account update activities for security and audit purposes.

3.2.4 UC004 Delete Account

Use Case ID:	UC004		
Use Case Name:	Delete Account		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The Supervisor removes an existing user account from the system. This operation permanently deletes all associated information for the selected account.	
Preconditions:	The Supervisor must be logged into the system with sufficient privileges. The account to be deleted must exist in the system.	
Postconditions:	The account is permanently removed from the system. A confirmation message is displayed to the Supervisor.	
Priority:	Medium	
Frequency of Use:	Occasional	
Normal Course of Events:	Actor	System
	1. Logs into the system.	1. Authenticates the Supervisor's credentials.
	2. Navigates to the "Account Management" section.	2. Displays a list of existing accounts.
	3. Selects the account to delete.	3. Displays the account details.
	4. Confirms the deletion.	4. Validates the request and deletes the account from the database.
	5. Receives a confirmation message of successful deletion.	5. Logs the deletion activity for auditing purposes.
Alternative Courses:	If the Supervisor cancels the deletion process after the warning message, the system aborts the operation and returns to the "Home" page.	

Exceptions:	The system fails to delete the account due to a database issue or dependency, in which case an error message is displayed.
Includes:	Validation of delete request.
Notes and Issues:	Include audit logging to track the deletion activity for security purposes.

3.2.5 UC005 Create Guest

Use Case ID:	UC005		
Use Case Name:	Create Guest		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ameerul Haqeem Bin Azlee
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The supervisor creates a guest profile in the Key Management System (KMS). The guest profile includes personal details required for key assignments and system tracking.	
Preconditions:	The supervisor must be logged into the system with valid credentials. The Key Management System must be online and operational.	
Postconditions:	The guest information is successfully created and stored in database. The system confirms the creation and provides a unique guest ID.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Logs into the system.	1. Validates the login credentials.
	2. Selects the “Create Guest” option.	2. Display guest form.
	3. Enters guest information.	3. Validates the entered guest information.
	4. Confirm the information	4. Create the guest information and create new guest ID.

	5. Receives confirmation of creation.	5. Display success message.
Alternative Courses:	If mandatory information is missing, the system prompts the supervisor to fill in the required fields before proceeding.	
Exceptions:	If the entered data violates validation rules (e.g., invalid date format), the system rejects the input and highlights the fields that need correction.	
Includes:	Validating data inputs.	
Notes and Issues:	Ensure data privacy regulations are met during handling the guest information.	

3.2.6 UC006 Update Guest

Use Case ID:	UC006		
Use Case Name:	Update Guest		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Ameerul Haqeem Bin Azlee
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	Supervisors update an existing guest profile in the Key Management System. This includes modifying personal information or key assignment details when necessary.	
Preconditions:	The supervisor must be logged into the system with valid credentials. The guest profile must exist in the database.	
Postconditions:	The guest profile is updated in the database with the new information. The system confirms the successful update.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Logs into the system.	1. Validates the login credentials.
	2. Selects the "Update Guest" option.	2. Displays the list of existing guest profiles.
	3. Searches for the guest profile.	3. Fetches and displays the requested guest details.
	4. Edits the required guest information.	4. Validates the updated data.
	5. Confirms the changes.	5. Updates the guest profile in the database and stores the changes.
	6. Receives confirmation of the update.	

		6. Displays a success message confirming the update.
Alternative Courses:	If the guest profile cannot be found, the system notifies the supervisor and prompts them to recheck the search criteria.	
Exceptions:	If the updated data violates validation rules (e.g., invalid phone number format), the system rejects the input and highlights the fields requiring correction.	
Includes:	Searching for guest profiles.	
Notes and Issues:	Ensure the system maintains a log of all updates made to guest profiles, including timestamps and the supervisor's ID for creating report.	

3.2.7 UC007 Create Key Number

Use Case ID:	UC007		
Use Case Name:	Create Key Number		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	Supervisor creating a new key number within the Key Management System. This functionality is used to register keys and assign them to specific houses.	
Preconditions:	The house that will associate with the key must already exist.	
Postconditions:	The new key number is registered. The system confirms successful creation of the key number.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Selects the “Key List” option in home page. 2. Selects the “Add” option 3. Enter key details 4. Submit key creation	1. Displays the list of the keys. 2. Display the key creation form 3. Validates the entered data 4. Register the key in the database
Alternative Courses:	If the house associated with the key does not exist, the system notifies the supervisor and prompts them to create or update the required data first.	
Exceptions:	If the key number already exists, the system rejects the creation request and provides an error message.	

	If required fields are missing or invalid (e.g., no block assigned), the system highlights the errors and prompts for correction.
Includes:	Validation of key details.
Notes and Issues:	Ensure that key numbers are unique within the system. Provide an option to batch-create key numbers for efficiency in large-scale operations.

3.2.8 UC008 View Key Number

Use Case ID:	UC008		
Use Case Name:	View Key Number		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Ameerul Haqeem Bin Azlee
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	Supervisors can access the details of registered keys.	
Preconditions:	The system database must contain key numbers to display.	
Postconditions:	The actor can view the list of key numbers and their associated details. The system ensures data visibility is limited to the actor's role and access rights.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Select the “Key List” option	1. Displays a list of all key numbers.
	2. Select a specific key number for more details.	2. Displays detailed information about the selected key.
	3. Views the information.	3. Ensures the data is accurate and up to date.
Alternative Courses:	If no key numbers exist in the database, the system displays a message indicating that no records are available.	

Exceptions:	If the actor does not have sufficient privileges to view key numbers, the system denies access and provides an error message.
Includes:	Data validation and role-based access control
Notes and Issues:	Supervisors should have access to all key number details for managerial oversight.

3.2.9 UC009 Update Key Number

Use Case ID:	UC009		
Use Case Name:	Update Key Number		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The Supervisor update an existing key number in the Key Management System. This functionality is used when a key number needs to be modified due to administrative updates, corrections, or reassignment purposes.	
Preconditions:	Supervisor logged into the system with valid credentials. The key number that needs to be updated must already exist in the system.	
Postconditions:	The selected key number is successfully updated in the system.	
Priority:	Medium	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Supervisor click “Update” button. 2. Supervisor edits the key number.	1.1.System validates the updated information. 2.1.If valid, the system updates the key number in the database.

	3. Supervisor update the information by click the “Save” button.	3.1.The system displays a confirmation message.
Alternative Courses:	If the key number entered is not valid or does not exist, the system displays an error message: "Key number not found. Please check and try again." If the Supervisor leaves any required fields blank, the system displays an error message prompting the Supervisor to complete them. If the system detects that the updated key number already exists, it displays an error message: "Duplicate key number. Please enter a unique key number."	
Exceptions:	3.1 If the system encounters a technical error (e.g., database connection failure), it displays a message: "Unable to update the key number at this time. Please try again later."	
Includes:	Validate User Input (a common sub-use case for validating form data).	
Notes and Issues:	Ensure the Supervisor can see a change log or history of updates for auditing purposes.	

3.2.10 UC010 Delete Key Number

Use Case ID:	UC010		
Use Case Name:	Delete Key Number		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ameerul Haqeem Bin Azlee
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	Supervisor has an authorized to remove a key number from the Key Management System.	
Preconditions:	The system database must contain the key number to be deleted. The key number being deleted must not currently be assigned to a guest or maintenance task.	
Postconditions:	The specified key number is successfully deleted from the system. The system updates record and ensures the deleted key number is no longer visible or accessible.	
Priority:	Medium	
Frequency of Use:	Low	
Normal Course of Events:	Actor	System
	1. Navigates to the “Key List” section	1. Displays a list of key numbers.
	2. Selects a key number to delete.	2. Highlights the selected key number.
	3. Click on “Edit” button	3. Display a key management page.
	4. Deletes the key number.	4. Removes the key number from the database and updates records.
	5. Views confirmation message.	

		5. Displays a success message indicating the deletion is complete.
Alternative Courses:	If the actor cancels the deletion action, the system does not remove the key number and returns to the previous state.	
Exceptions:	If the actor does not have sufficient privileges to delete key numbers, the system denies access and provides an error message.	
Includes:	Role-based access control.	
Notes and Issues:	Supervisors must ensure that key numbers being deleted are no longer in use or relevant. Maintain an audit log of deleted key numbers for record-keeping and traceability.	

3.2.11 UC011 Create Assign Key

Use Case ID:	UC011		
Use Case Name:	Create Assign Key		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The supervisor is responsible for assigning keys to the staff through the Key Management System to ensure proper tracking and management.	
Preconditions:	The system must have key numbers available in the database. Staff information must exist in the system.	
Postconditions:	The key number is successfully assigned to the selected staff. The system updates the status of the assigned key based on the role assigned.	
Priority:	High	
Frequency of Use:	Moderate	
Normal Course of Events:	Actor	System
	1. Navigates to the "Key Page" section. 2. Select the key number. 3. Selects staff member. 4. Confirms the assignment.	1. Displays a list of keys. 2. Display a key page. 3. Highlights the selected individual. 4. Update the key status to inspection or maintenance.
Alternative Courses:	If the actor cancels the assignment action, the system does not assign the key and returns to the previous state.	

Exceptions:	If a system error occurs during the assignment process, the system displays a message indicating the action failed and logs the issue.
Includes:	Validation of available keys.
Notes and Issues:	Consider adding a notification feature to inform the assigned guest or staff member. Ensure that assigned keys are correctly tracked in the system to prevent duplicate assignments.

3.2.12 UC012 View Assign Key

Use Case ID:	UC012		
Use Case Name:	View Assign Key		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Staff, Inspector, Supervisor and Manager	
Description:	Allows authorized users to view the list of keys that have been assigned to them.	
Preconditions:	Supervisor has the necessary permissions and credentials to logged into the system and successfully authenticated to the system. The system is operational and accessible.	
Postconditions:	The list and details of keys that been assigned to the authorized staff and inspector All staff, inspector, supervisor and manager can view the assigned key.	
Priority:	High	
Frequency of Use:	Medium	
Normal Course of Events:	Actor	System
	1. Staff, Inspector, Supervisor and Manager logs into the system 2. User select view assigned keys.	1.1 Authenticates user credentials 2.1 System displays the list assigned keys information
Alternative Courses:	If supervisor cannot view the assigned keys, it may unknowingly increase security risks	
Exceptions:	1.1 Unauthorized access	

	2.1 Only necessary information of the authorized staff will be displayed
Includes:	Key Identification – Identifiers for each key such as key ID Assignment Date – Date and time when the key is scheduled. Expiry Date – Date and time on when the key is supposed to return.
Notes and Issues:	The key will be stored securely to prevent any unauthorized access or modification

3.2.13 UC013 Update Assign Key

Use Case ID:	UC013		
Use Case Name:	Update Assign Key		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The supervisor can reassign a key to a different individual or update details related to the assigned key as needed.	
Preconditions:	The key must already be assigned in the system.	
Postconditions:	The assigned key details are successfully updated in the system.	
Priority:	Medium	
Frequency of Use:	Low	
Normal Course of Events:	Actor	System
	1. Navigates to the "Assignment" section.	1. Displays a list of all assigned keys and their details.
	2. Selects an assigned key	2. Highlights the selected key and displays its details.
	3. Updates the key's assignment details	3. Validates the input data.
	4. Confirms the update.	4. Saves the updated key assignment.
	5. Views confirmation message.	

		5. Displays a success message indicating the update is complete.
Alternative Courses:	If no valid data is provided, the system prompts the actor to enter correct details.	
Exceptions:	If the actor does not have sufficient privileges to update assigned keys, the system denies access and provides an error message. If a system error occurs during the update process, the system displays a failure message and logs the issue.	
Includes:	Validation of key assignment data.	
Notes and Issues:	• Provide the actor with an option to notify the new assigned individual (e.g., guest or staff) about the update. • Add safeguards to prevent accidental deletion or overwriting of key details during the update process.	

3.2.14 UC014 Delete Assign Key

Use Case ID:	UC014		
Use Case Name:	Delete Assign Key		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The Supervisor removes an existing key assignment from the system. This operation unassigns a key from its current holder and updates the key's status to "Available."	
Preconditions:	The Supervisor must be logged into the system with sufficient privileges. The key assignment to be deleted must exist in the system.	
Postconditions:	A confirmation message is displayed to the Supervisor.	
Priority:	Medium	
Frequency of Use:	Occasional	
Normal Course of Events:	Actor	System
	1. Navigates to the "Assignment" section. 2. Selects the specific key assignment to delete. 3. Delete the key's assignment details 4. Confirms the deletion.	1. Displays a list of all assigned keys and their details. 2. Displays a warning message asking for confirmation. 3. Validates the input data.

	5. Views confirmation message.	4. Validates the request and unassigns the key. 5. Displays a success message indicating the deletion is complete.
Alternative Courses:	If the Supervisor cancels the deletion process after the warning message, the system aborts the operation and returns to the "Home" page.	
Exceptions:	The system fails to delete the assignment due to a database issue, in which case an error message is displayed.	
Includes:	Validation of deletion request.	
Notes and Issues:	Prevent deletion of assignments if the key is currently under maintenance or associated with unresolved incidents.	

3.2.15 UC015 Create Assignment

Use Case ID:	UC015		
Use Case Name:	Create Assignment		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ameerul Haqeem Bin Azlee
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The Supervisor assigns a key to a specific user, such as a inspector or maintenance staff, updating the system to reflect the key's assigned status.	
Preconditions:	The Supervisor must be logged into the system with appropriate privileges. The key to be assigned must be in "Checked Out" status. The user receiving the assignment must have a valid profile in the system.	
Postconditions:	The key is marked as "Assigned" in the system. The assignment record is stored in the database.	
Priority:	High	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Key List" section. 2. Selects the key to assign and the user profile to whom the key will be assigned. 3. Selects the key to assign.	1. Displays a list of available keys. 2. Displays a warning message asking for confirmation.

		3. Validates the selection and displays a summary for confirmation.
	4. Confirms the assignment.	4. Updates the key's status to "Assigned" and links it to the selected user.
	5. Receives a confirmation message of successful assignment.	5. Logs the assignment activity for auditing purposes.
Alternative Courses:	If the Supervisor cancels the assignment process before confirmation, the system aborts the operation and returns to the "Home" page.	
Exceptions:	The system displays an error message if the key is no longer available or the user profile is invalid..	
Includes:	Validation of key availability.	
Notes and Issues:	Ensure the system prevents duplicate assignments of the same key.	

3.2.16 UC016 View Assignment

Use Case ID:	UC016		
Use Case Name:	View Assignment		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor, Inspector, Manager	
Description:	The user retrieves and views details of key assignments, including information about assigned keys, users, and assignment status.	
Preconditions:	The system contains existing key assignment records.	
Postconditions:	Key assignment details are displayed to the user.	
Priority:	Medium	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Assignment" section.	1. Displays interface for assignments.
	2. Finds for assignments using specific criteria (e.g., key number, name, or assignment status).	2. Retrieves and displays matching assignment records.
	3. Selects a specific assignment from the list to view detailed information.	3. Displays detailed information about the selected assignment,

		including key number, assigned user, assignment date, and status.
Alternative Courses:	If no assignments match the search criteria, the system displays a message indicating "No records found.	
Exceptions:	The system shows an error message if it encounters a database connectivity issue while retrieving records.	
Includes:	Display of detailed assignment records.	
Notes and Issues:	Provide options to export or print assignment details for reporting purposes.	

3.2.13 UC017 Update Assignment

Use Case ID:	UC017		
Use Case Name:	Update Assignment		
Created By:	Ahmad Emir Erfan Bin Mohd Zubri	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The supervisor updates an existing key assignment by selecting a new key number and/or reassigning the key to a different staff member using dropdown menus.	
Preconditions:	The system contains existing assignment records.	
Postconditions:	The updated assignment is saved in the system. The new key number and/or staff member information is reflected in the assignment record.	
Priority:	High	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Assignment" section. 2. Selects the assignment to be updated. 3. Uses the dropdown menu to select a new key number.	1. Displays interface for assignments. 2. Opens the assignment details page with current key number and staff details. 3. Updates the key number field in the assignment.

	4. Uses the dropdown menu to select a new staff member. 5. Clicks the "Save" button to confirm changes.	4. Updates the staff field in the assignment. 5.1 Validates the updates and saves the changes to the database. 5.2 Displays a confirmation message indicating successful update.
Alternative Courses:	If the supervisor cancels the operation before saving, the system discards any changes and returns to the assignment list.	
Exceptions:	If the database is unavailable during the update process, the system notifies the user and halts the update.	
Includes:	Dropdown selection for key numbers. Dropdown selection for staff members.	
Notes and Issues:	Ensure dropdowns display only valid, unassigned key numbers and eligible staff members. Provide a user-friendly interface with clear instructions for updating assignments.	

3.2.18 UC018 Generate Report

Use Case ID:	UC018		
Use Case Name:	Generate Report		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor	
Description:	The user generates a report detailing system data such as key assignments, staff activity, and maintenance logs for review or auditing purposes.	
Preconditions:	The system must have sufficient data to generate a meaningful report.	
Postconditions:	A report is generated and made available for viewing, downloading, or printing.	
Priority:	High	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Reports" section in the main menu. 2. Selects the desired report type (e.g., key assignments, staff activity, or maintenance logs). 3. Clicks the "Generate Report" button.	1. Displays the report generation interface 2. Updates the interface to display filters relevant to the selected report type. 3. Processes the data and generates the report in the

		chosen format (e.g., PDF or Excel).
Alternative Courses:	If the actor does not apply filters, the system generates a report with all available data. If the actor cancels the operation, the system discards any selected filters and returns to the main report menu.	
Exceptions:	If no data matches the selected filters, the system displays a message indicating no results. If the system encounters an error during report generation, it notifies the user and suggests retrying.	
Includes:	Filter selection for report customization. Format selection (e.g., PDF, Excel).	
Notes and Issues:	Ensure reports are formatted professionally for easy interpretation.	

3.2.19 UC019 View Report

Use Case ID:	UC019		
Use Case Name:	View Report		
Created By:	Ameerul Haqeem Bin Azlee	Last Updated By:	Muhammad Nursyaziq Raihan Bin Saian
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor, Manager	
Description:	The user views a previously generated report directly within the system interface for review or analysis.	
Preconditions:	The user must be logged into the system with the appropriate privileges. At least one report must have been generated and stored in the system.	
Postconditions:	The selected report is displayed for the user.	
Priority:	Medium	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Reports" section in the main menu. 2. Selects a report from the list. 3. Reviews the report on-screen.	1. Displays the report generation interface 2. Loads and displays the selected report in a readable format.

		3. Provides options for zooming, scrolling, and navigating through pages.
Alternative Courses:	If the actor searches for a specific report using filters (e.g., date range, report type), the system narrows down the list of available reports.	
Exceptions:	If no reports are available, the system displays a message indicating no results. If the selected report cannot be loaded due to an error, the system notifies the user and suggests retrying.	
Includes:	Report searches and filtering.	
Notes and Issues:	Ensure reports are displayed in a user-friendly layout for on-screen reading.	

3.2.20 UC019 Update Report

Use Case ID:	UC020		
Use Case Name:	View Report		
Created By:	Muhammad Nursyaziq Raihan Bin Saian	Last Updated By:	Ahmad Emir Erfan Bin Mohd Zubri
Date Created:	1/1/2025	Date Last Updated:	7/1/2025

Actor:	Supervisor, Manager	
Description:	The user views a previously generated report directly within the system interface for review or analysis.	
Preconditions:	The user must be logged into the system with the appropriate privileges. At least one report must have been generated and stored in the system.	
Postconditions:	The selected report is displayed for the user.	
Priority:	Medium	
Frequency of Use:	Regular	
Normal Course of Events:	Actor	System
	1. Navigates to the "Reports" section in the main menu. 2. Selects a report from the list. 3. Reviews the report on-screen.	1. Displays the report generation interface 2. Loads and displays the selected report in a readable format. 3. Provides options for zooming, scrolling, and navigating through pages.
Alternative Courses:	If the actor searches for a specific report using filters (e.g., date range, report type), the system narrows down the list of available reports.	
Exceptions:	If no reports are available, the system displays a message indicating no results. If the selected report cannot be loaded due to an error, the system notifies the user and suggests retrying.	
Includes:	Report searches and filtering.	
Notes and Issues:	Ensure reports are displayed in a user-friendly layout for on-screen reading.	

4. Other Non-functional Requirements

4.0 Performance Requirements

The system must provide a smooth and responsive interaction to ensure that the researchers can conduct the test, and the participants can perform the test efficiently. The system should respond to user actions such as button clicks, and page navigation.

Database operations including storing participants information, retrieving test question, and recording participants reaction must be complete within 2 seconds to avoid delays during the test. Reaction time recording and response must be processed accurately.

4.1 Safety and Security Requirements

The system must ensure the security and confidentiality of research data and participant information. All sensitive data, including participant records and experiment results, must be stored securely in the system database.

User authentication must be implemented to restrict system access only to authorized administrators and researchers. Passwords must follow strong security standards, including a combination of uppercase letters, lowercase letters, numbers, and special characters.

To prevent data loss, the system must implement automatic data saving during experiment sessions and ensure secure data transmission between the client and server. Access to participant data must be restricted according to user roles.

Additionally, system activities such as user logins, experiment sessions, and data modifications should be logged to maintain accountability and traceability.

4.2 Software Quality Attributes

4.2.1 Adaptability:

The system is designed to be adaptable so that additional stress-induction methods or test modules can be integrated in the future. The flexible system architecture allows researchers to create new experiment templates and configure parameters without major system changes.

4.2.2 Maintainability:

The system is developed with a modular structure that allows developers to easily maintain and update the software. Bug fixes, feature improvements, and security updates can be implemented efficiently without affecting other system components.

4.2.3 Portability:

The system is a web-based platform that can be accessed through modern web browsers which support major operating system such as Windows and macOS.

4.2.4 Reliability:

The system ensures reliable operation by securely storing experiment data and participant information. The database system is designed to maintain data integrity and reduce the risk of data loss during experiment sessions.

4.2.5 Usability:

The system uses a user-friendly interface designed for individuals with basic computer knowledge, ensuring ease of use. It is accessible for any devices with internet connection, providing convenience and flexibility for users to connect and use the system anywhere.

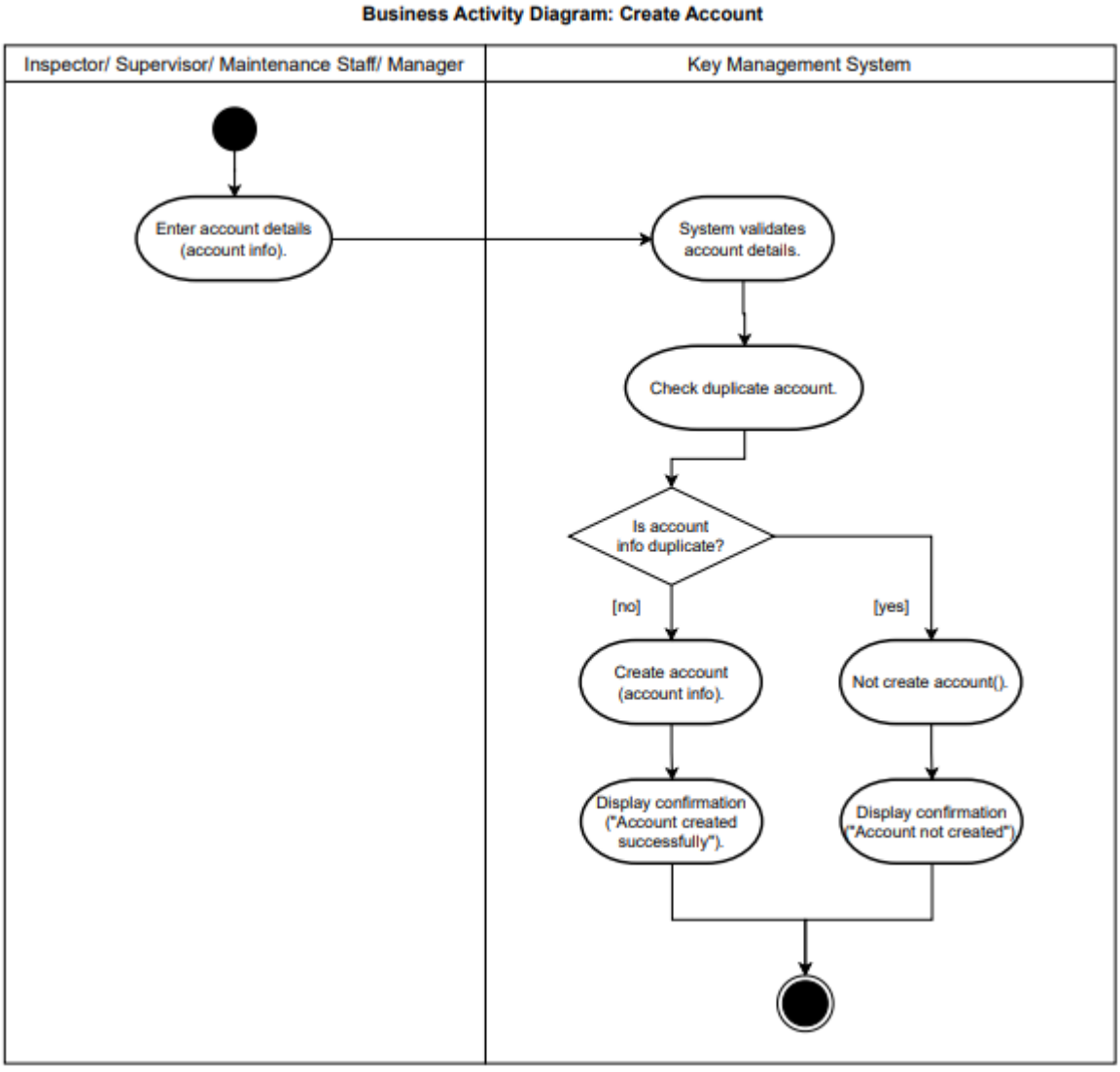
4.2.6 Performance Efficiency:

The system is optimized to provide efficient performance during experiment sessions. It ensures that stimuli presentation, response recording, and data processing occur quickly and accurately to support real-time experiment execution.

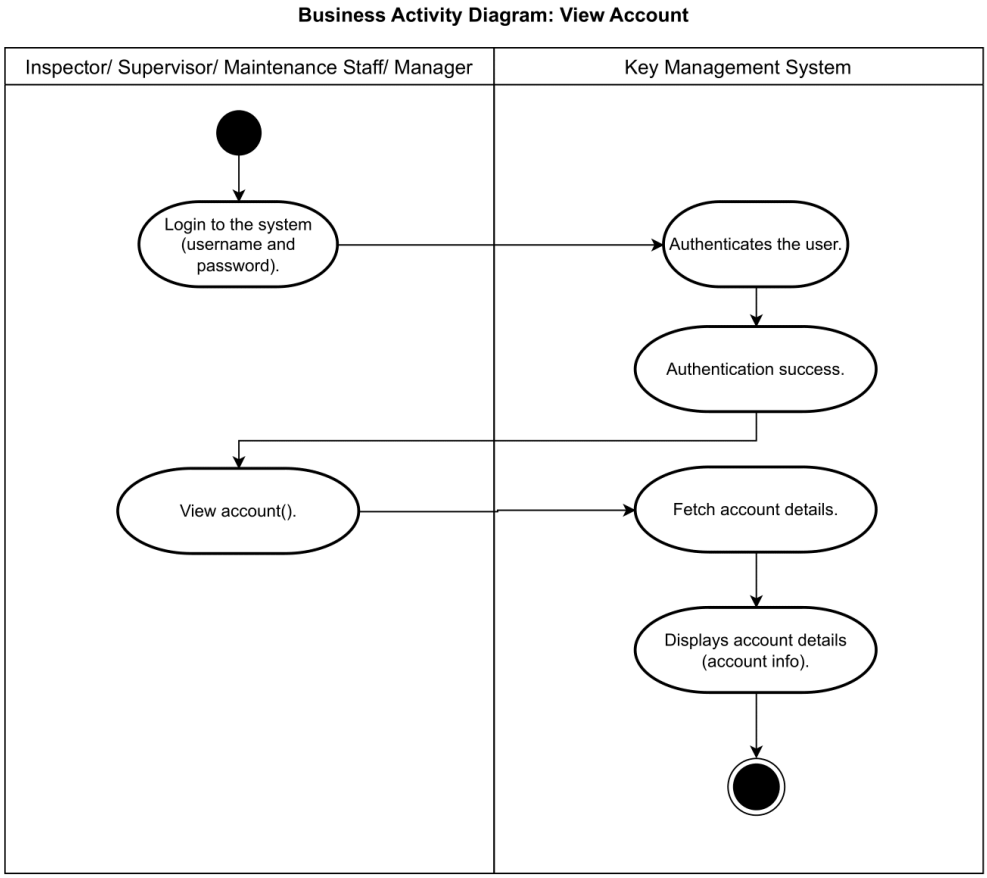
Appendix A – REQUIREMENT MODELS

1. BUSINESS ACTIVITY DIAGRAM

BAD001 Create Account

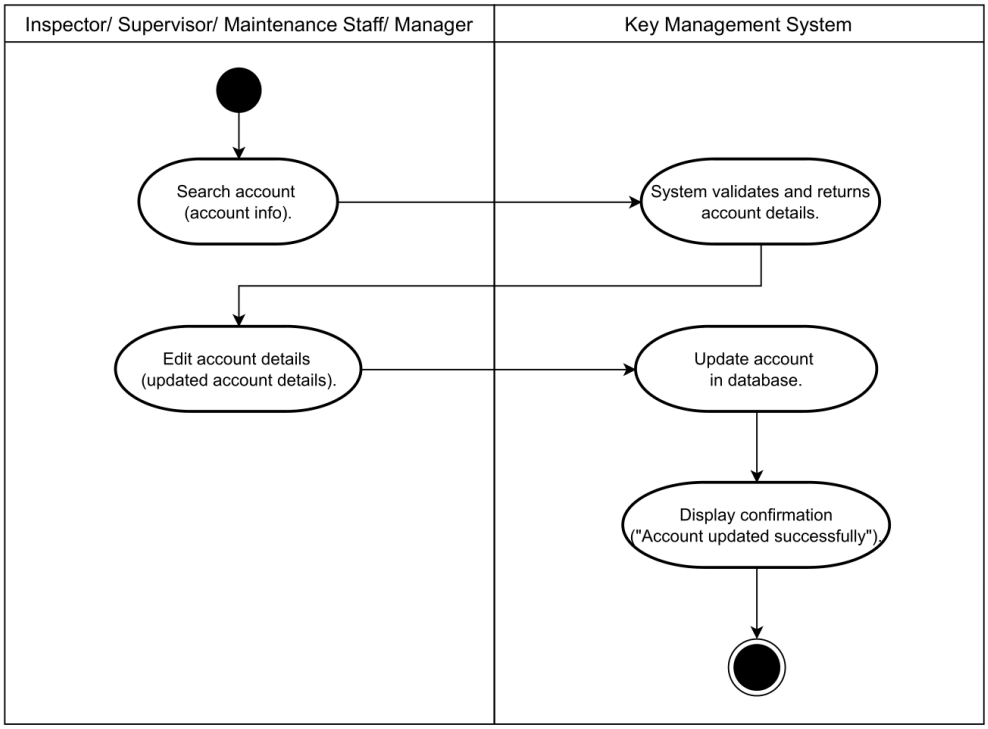


BAD002 View Account

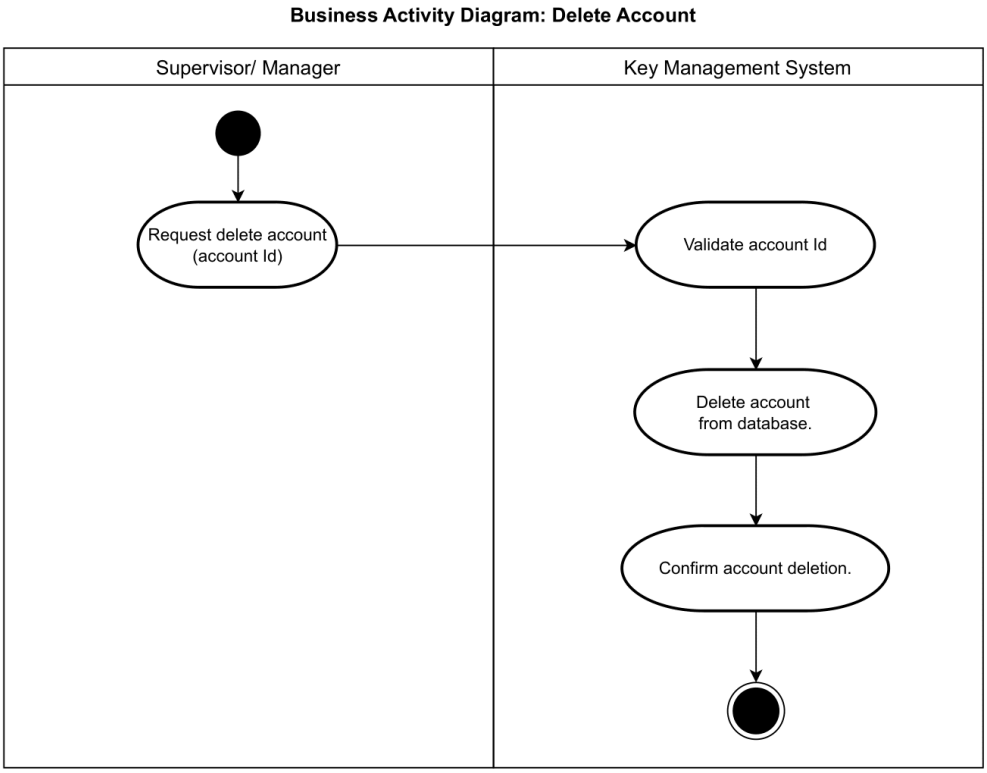


BAD003 Update Account

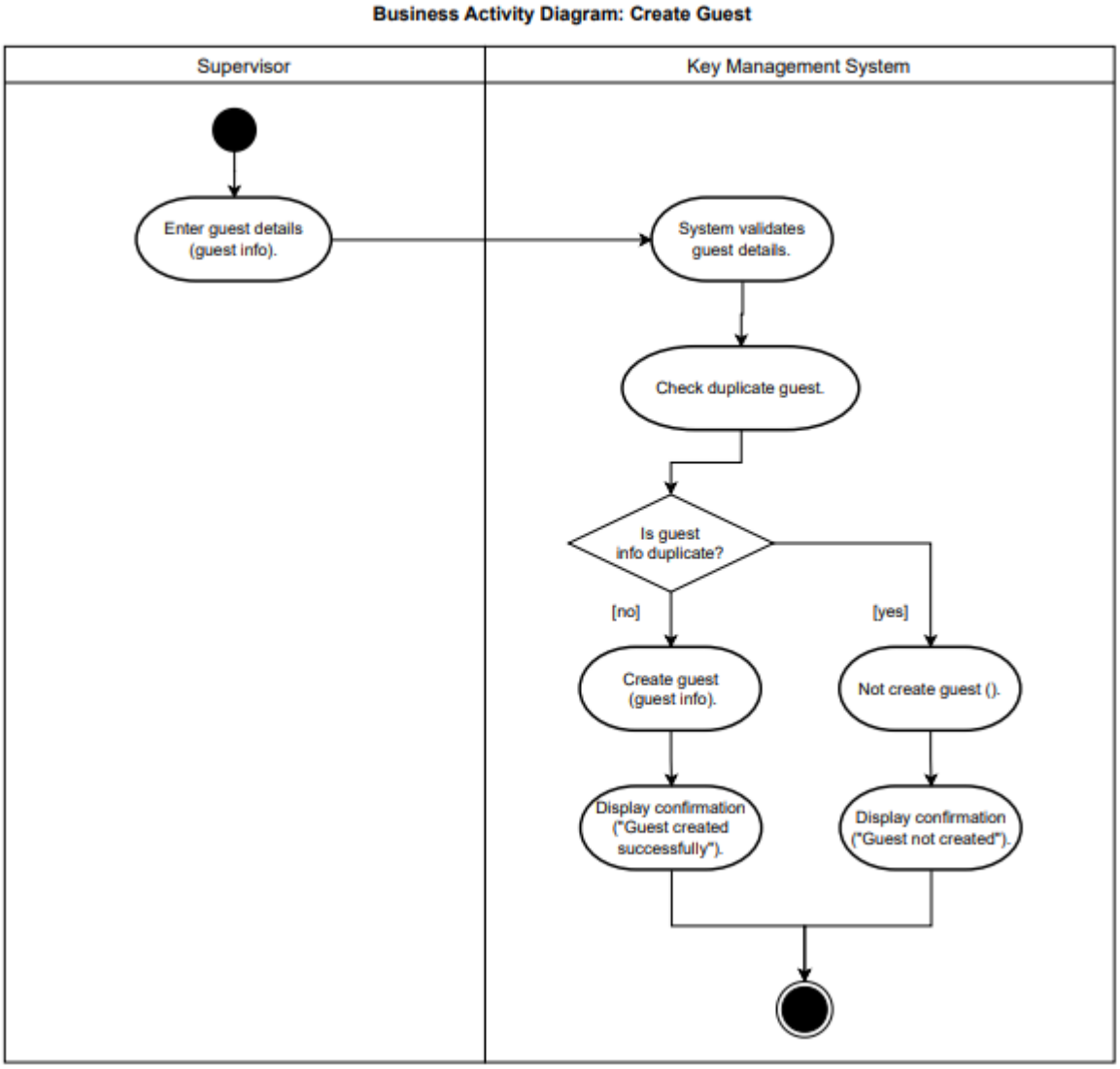
Business Activity Diagram: Update Account



BAD004 Delete Account

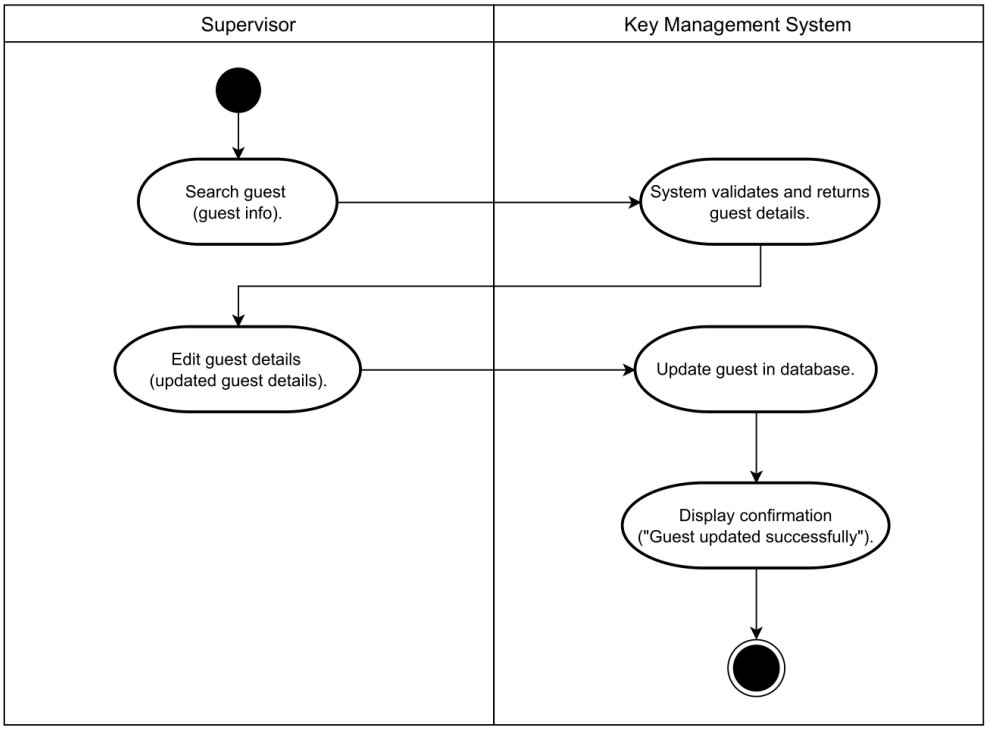


BAD005 Create Guest

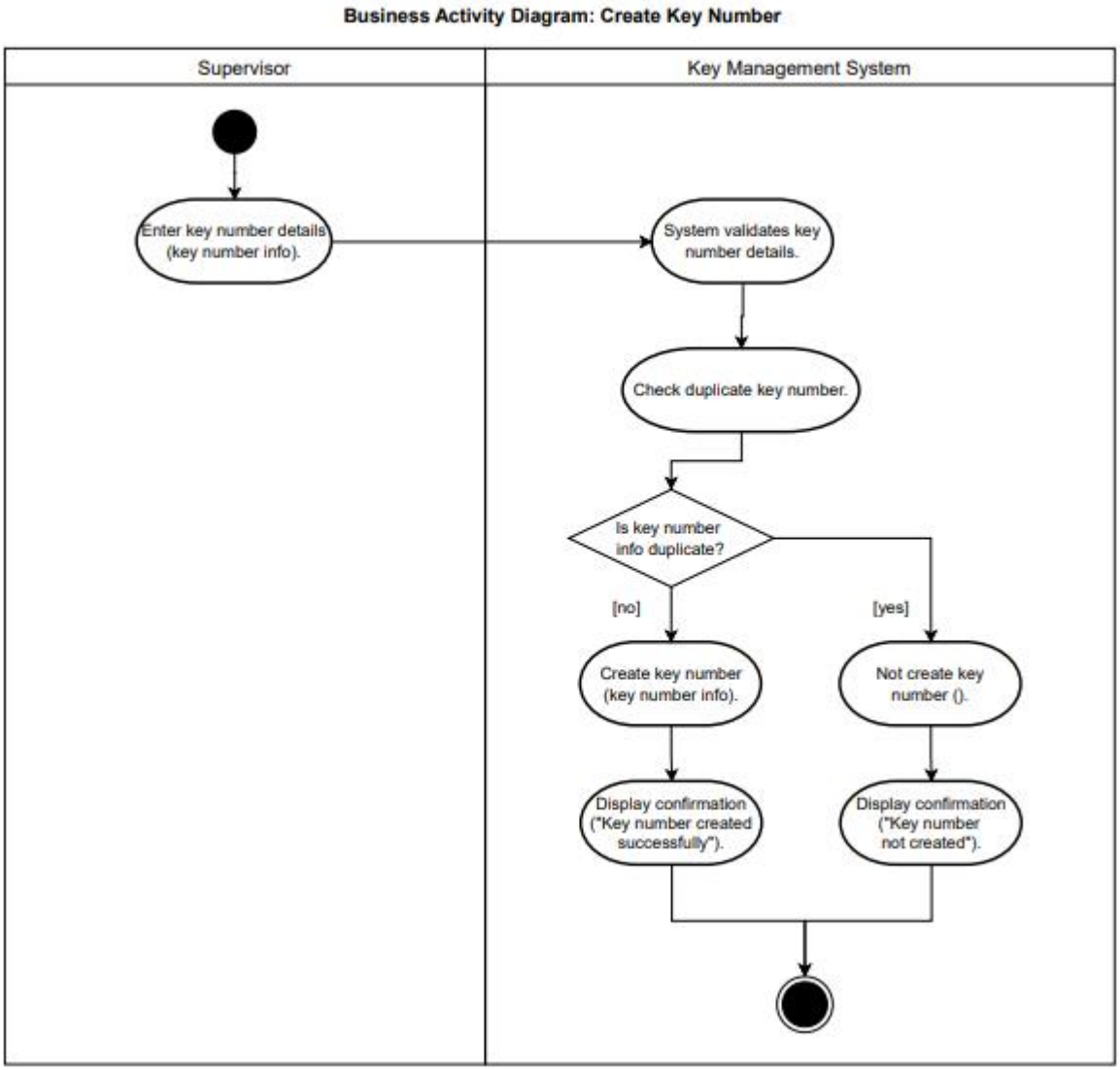


BAD006 Update Guest

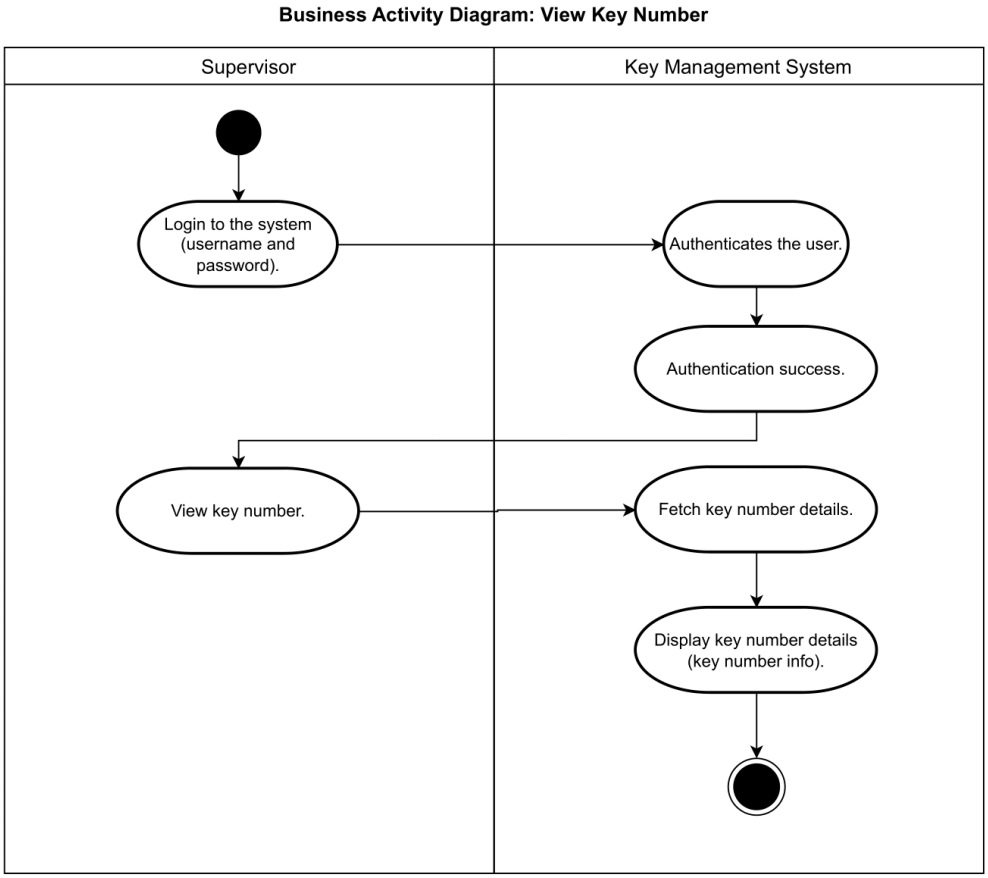
Business Activity Diagram: Update Guest



BAD007 Create Key Number

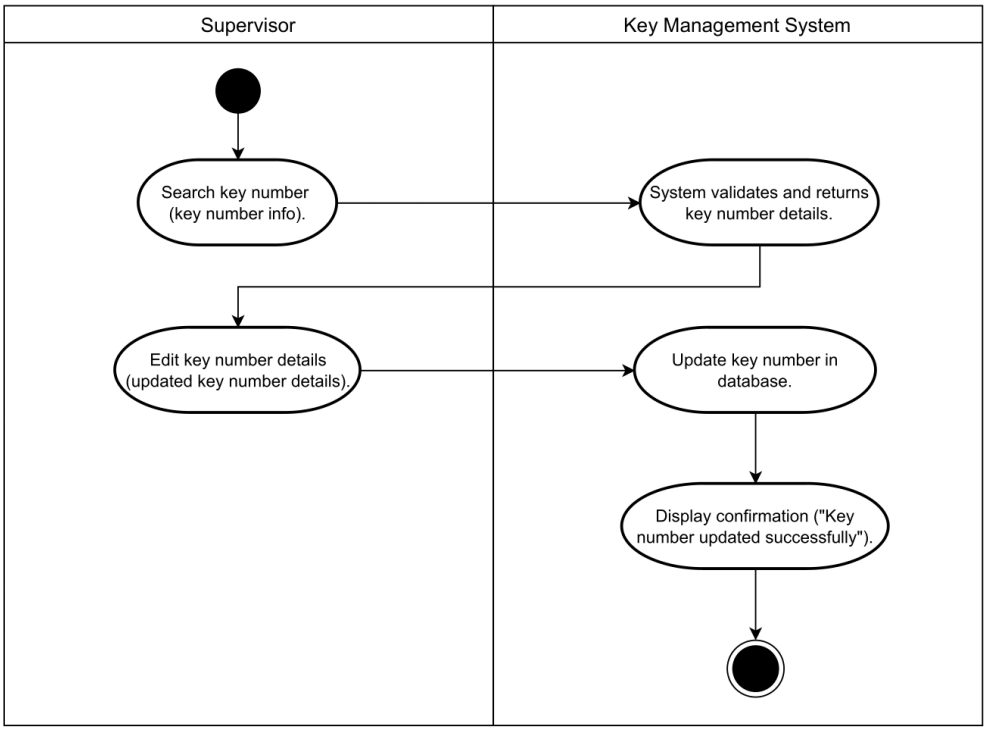


BAD008 View Key Number

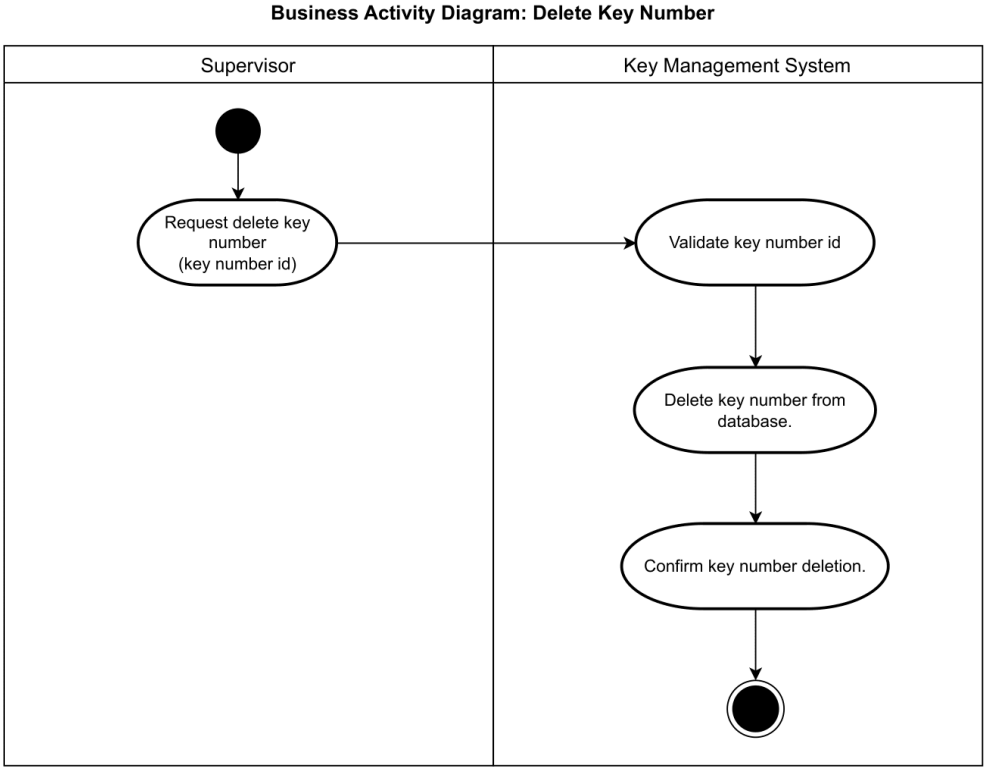


BAD009 Update Key Number

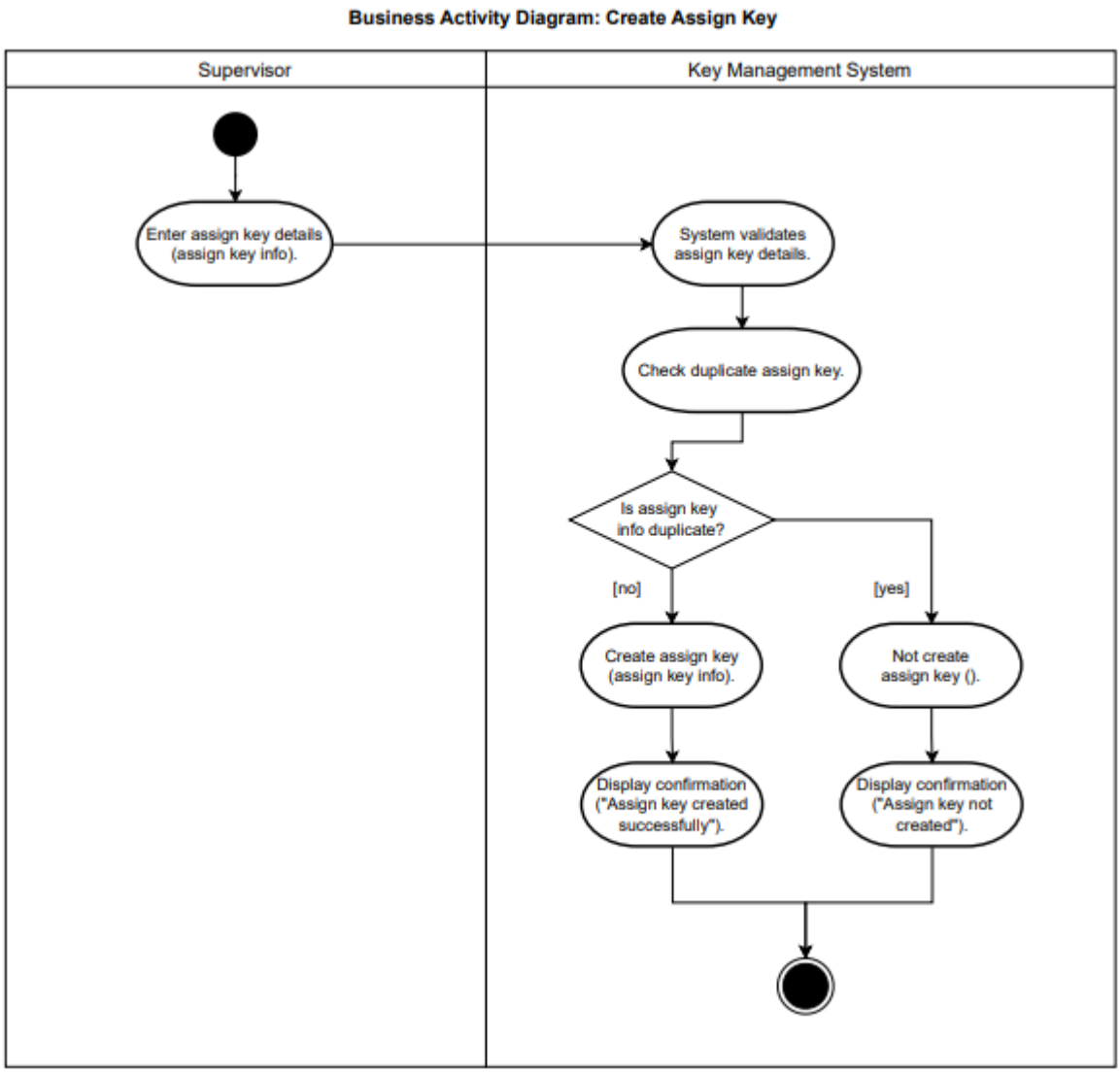
Business Activity Diagram: Update Key Number



BAD010 Delete Key Number

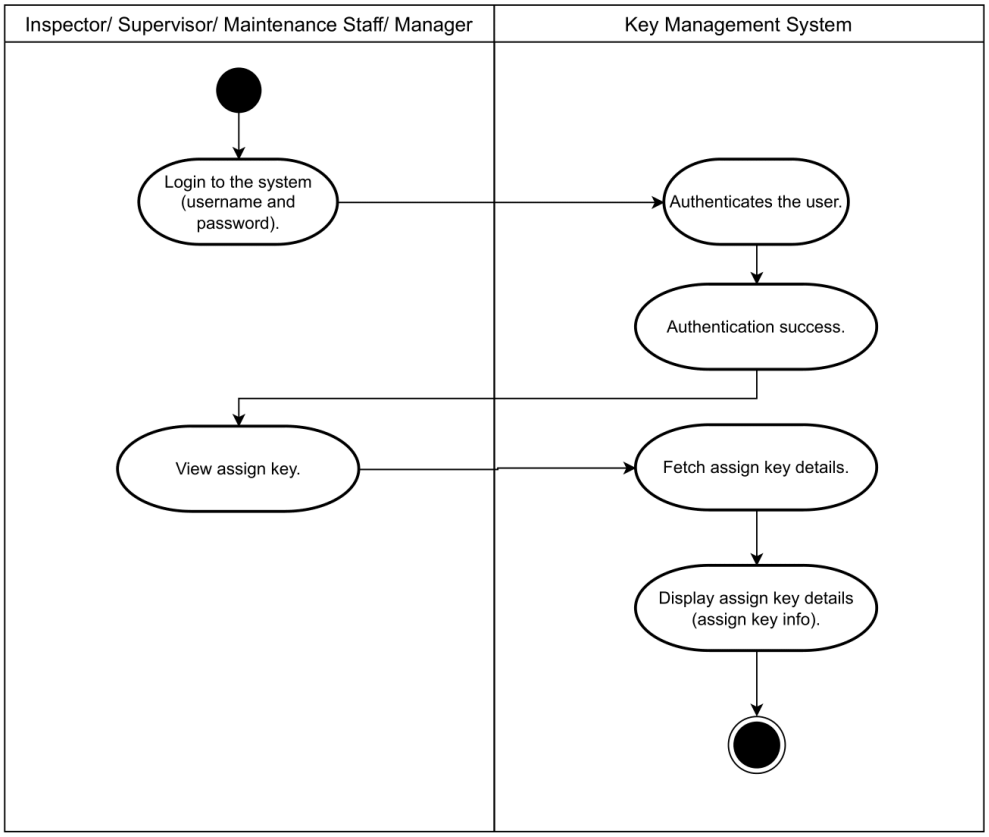


BAD011 Create Assign Key



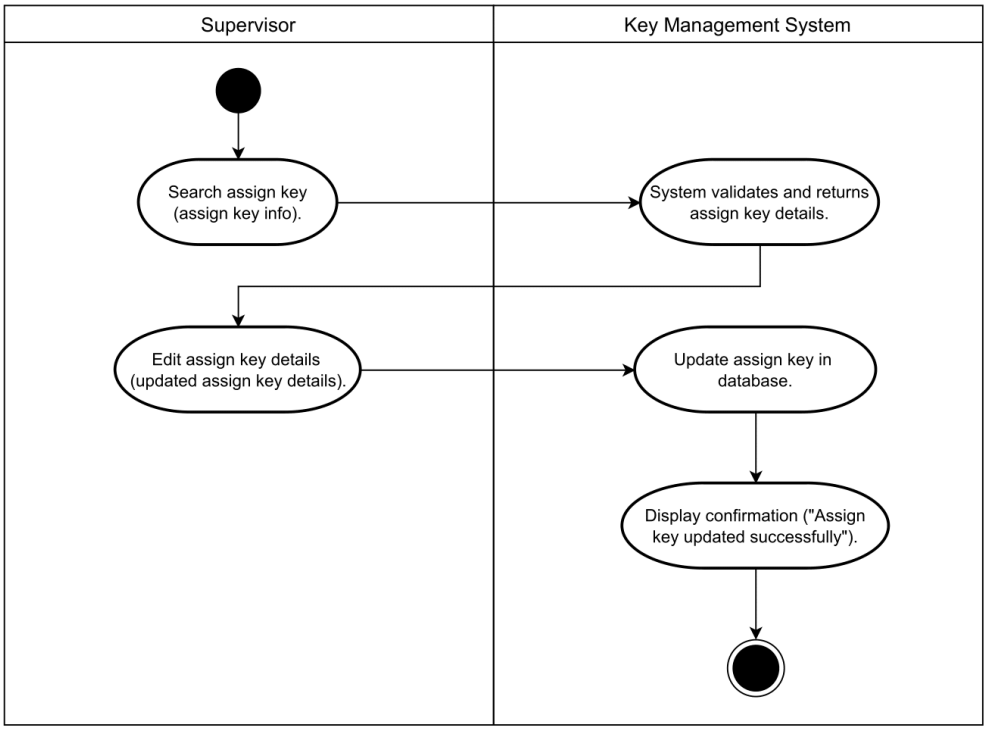
BAD012 View Assign Key

Business Activity Diagram: View Assign Key



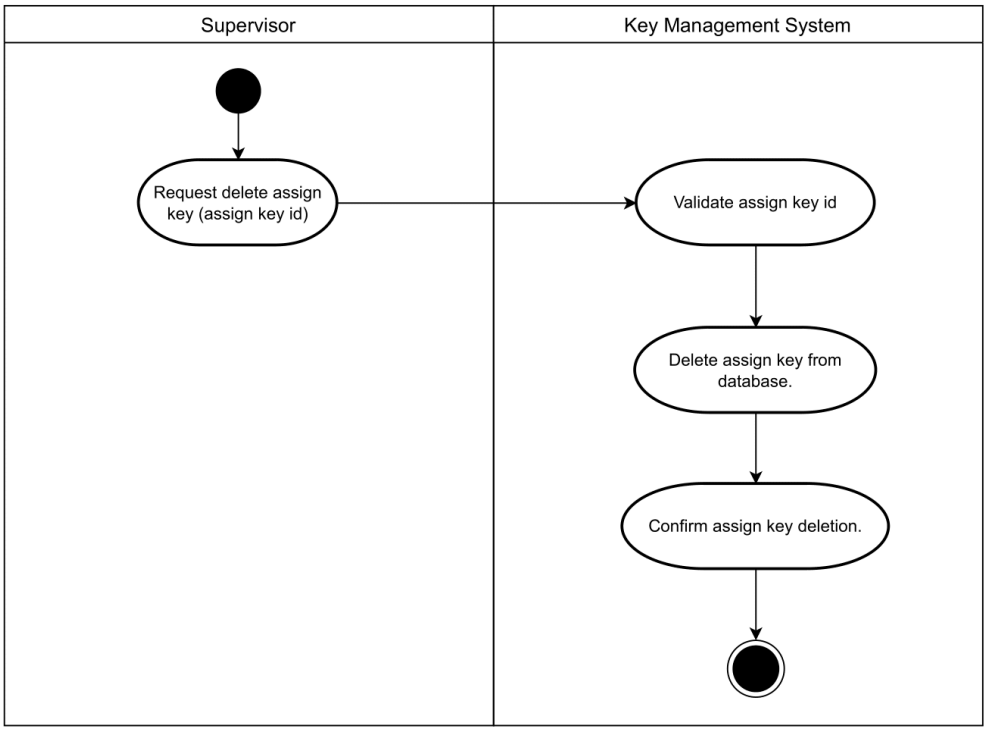
BAD013 Update Assign Key

Business Activity Diagram: Update Assign Key

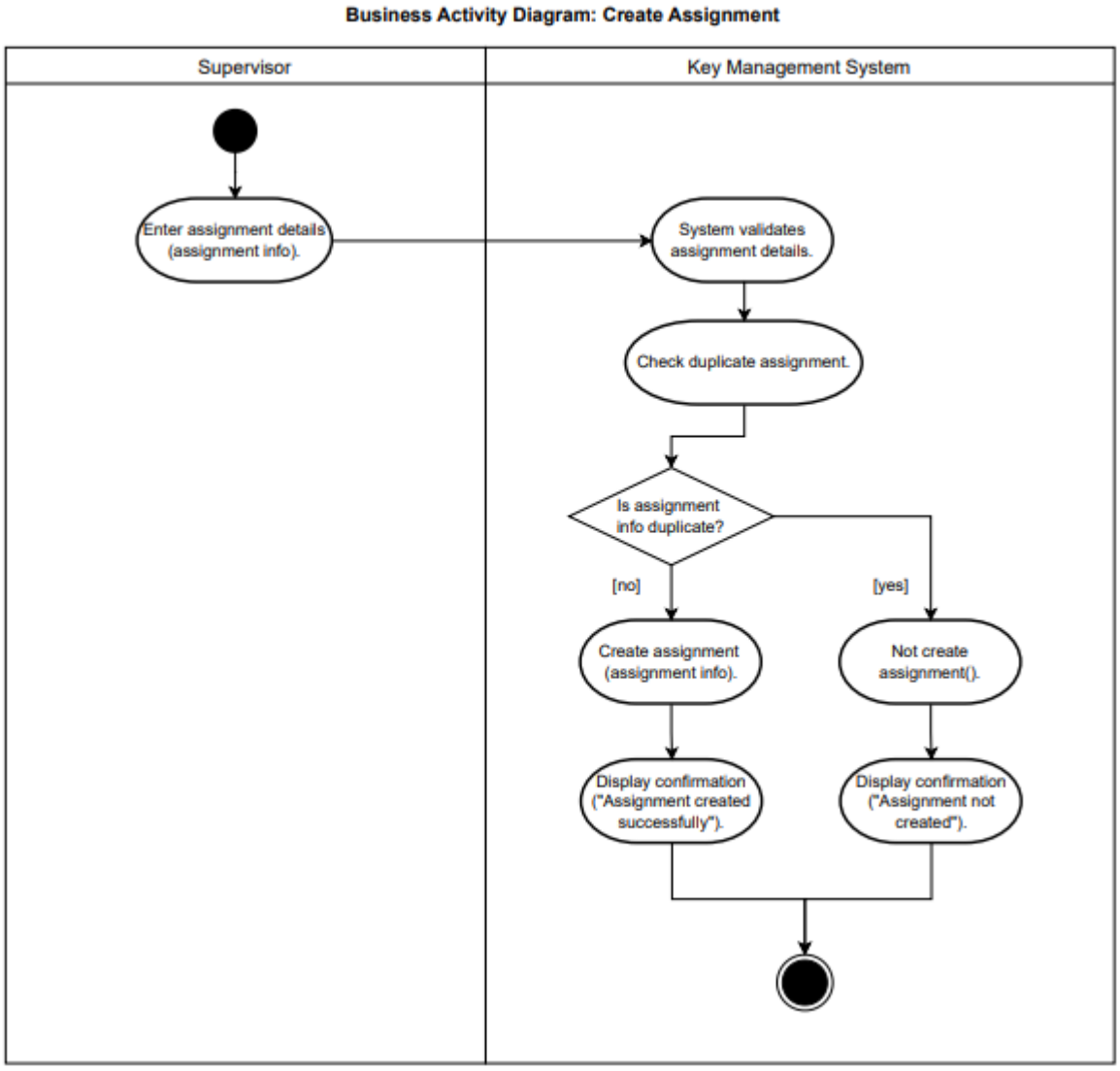


BAD014 Delete Assign Key

Business Activity Diagram: Delete Assign Key

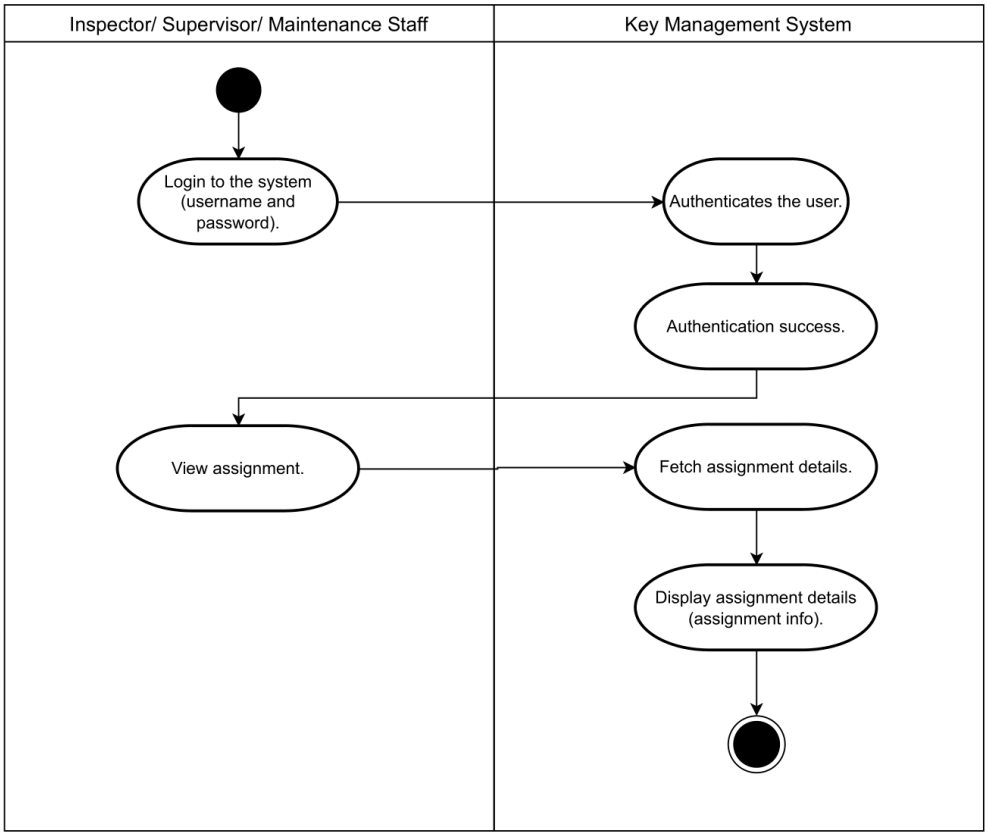


BAD015 Create Assignment



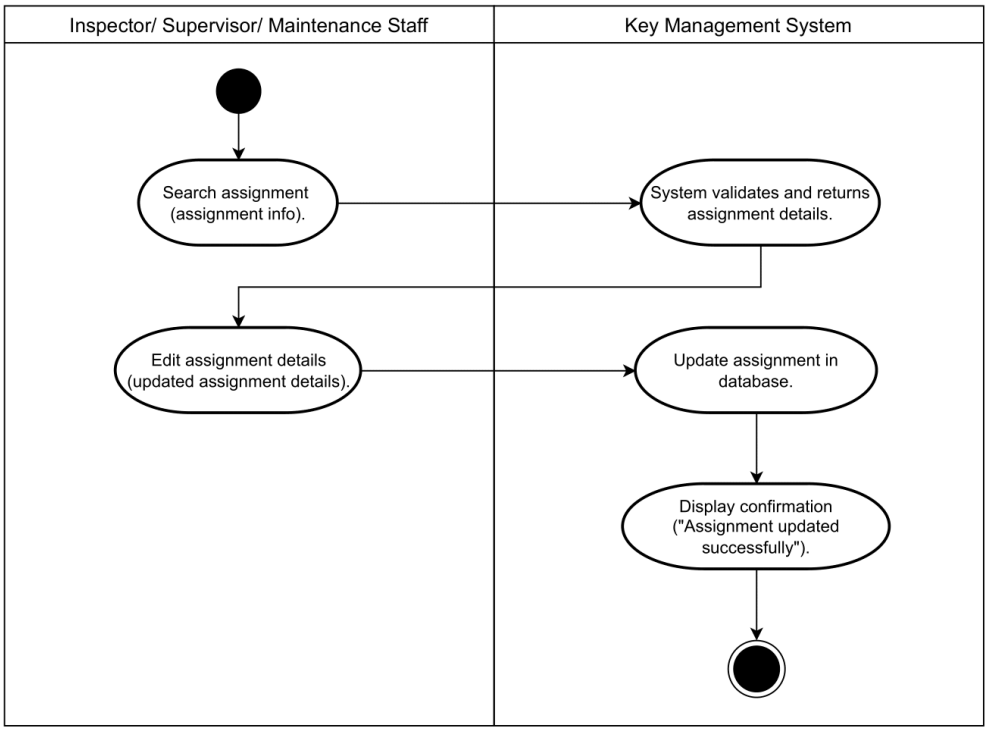
BAD016 View Assignment

Business Activity Diagram: View Assignment

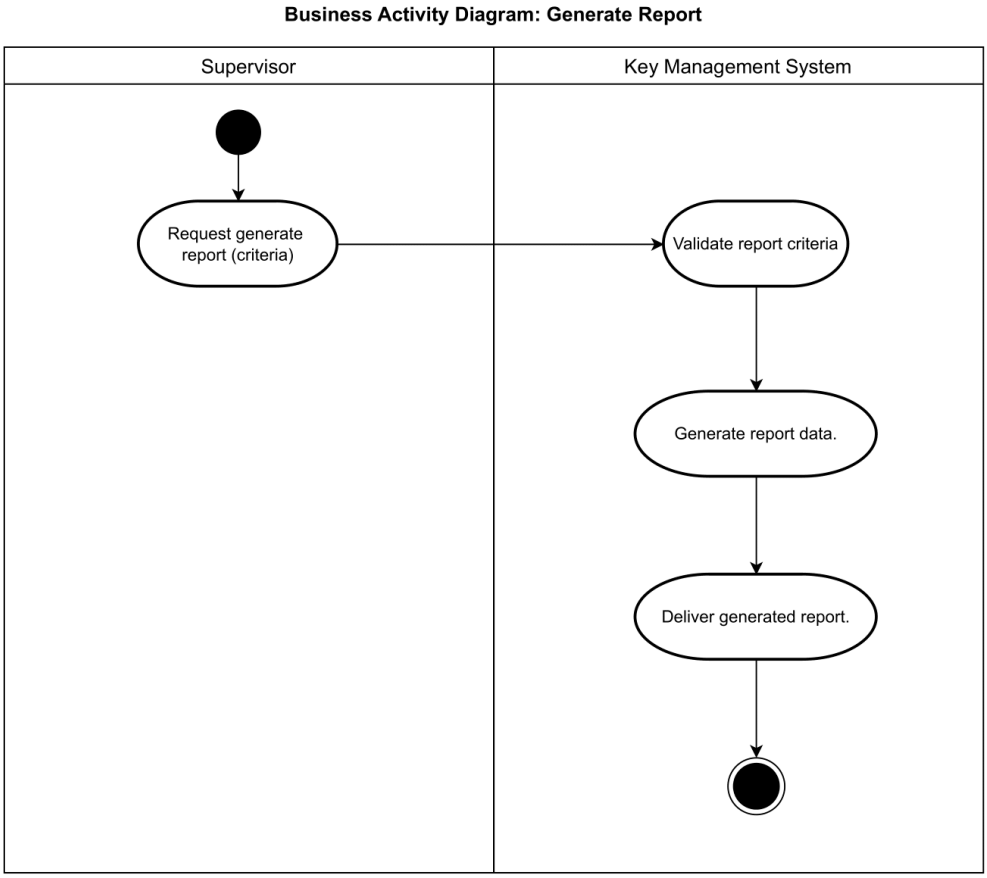


BAD017 Update Assignment

Business Activity Diagram: Update Assignment

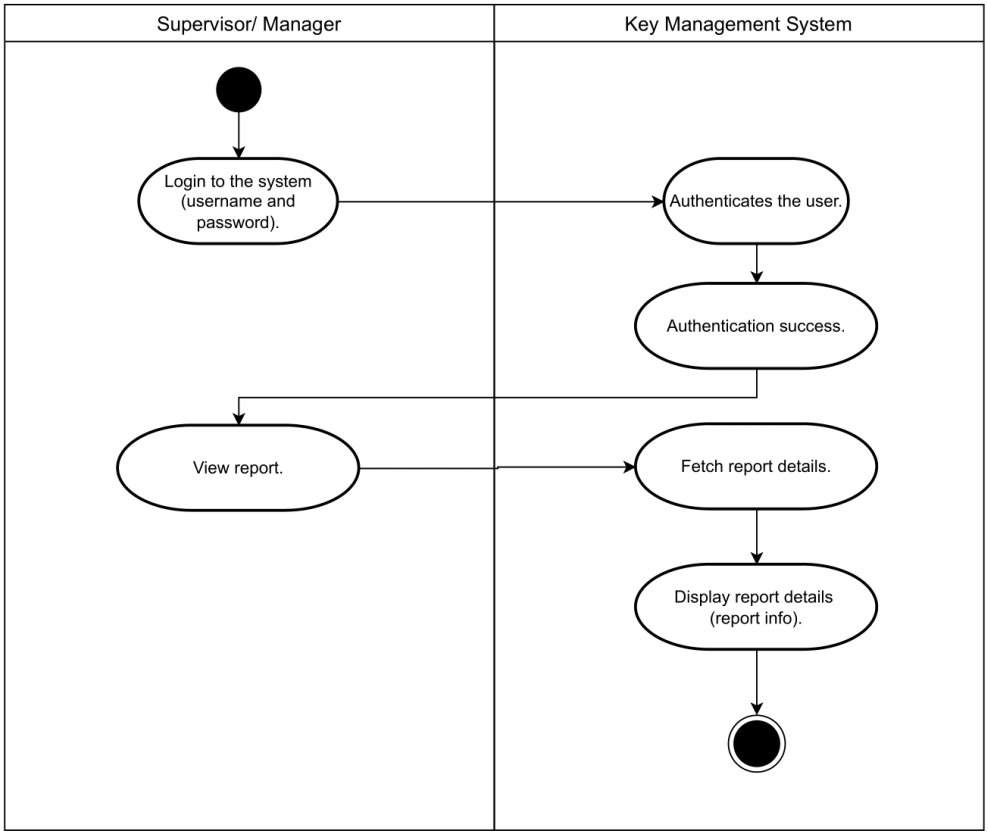


BAD018 Generate Report



BAD019 View Report

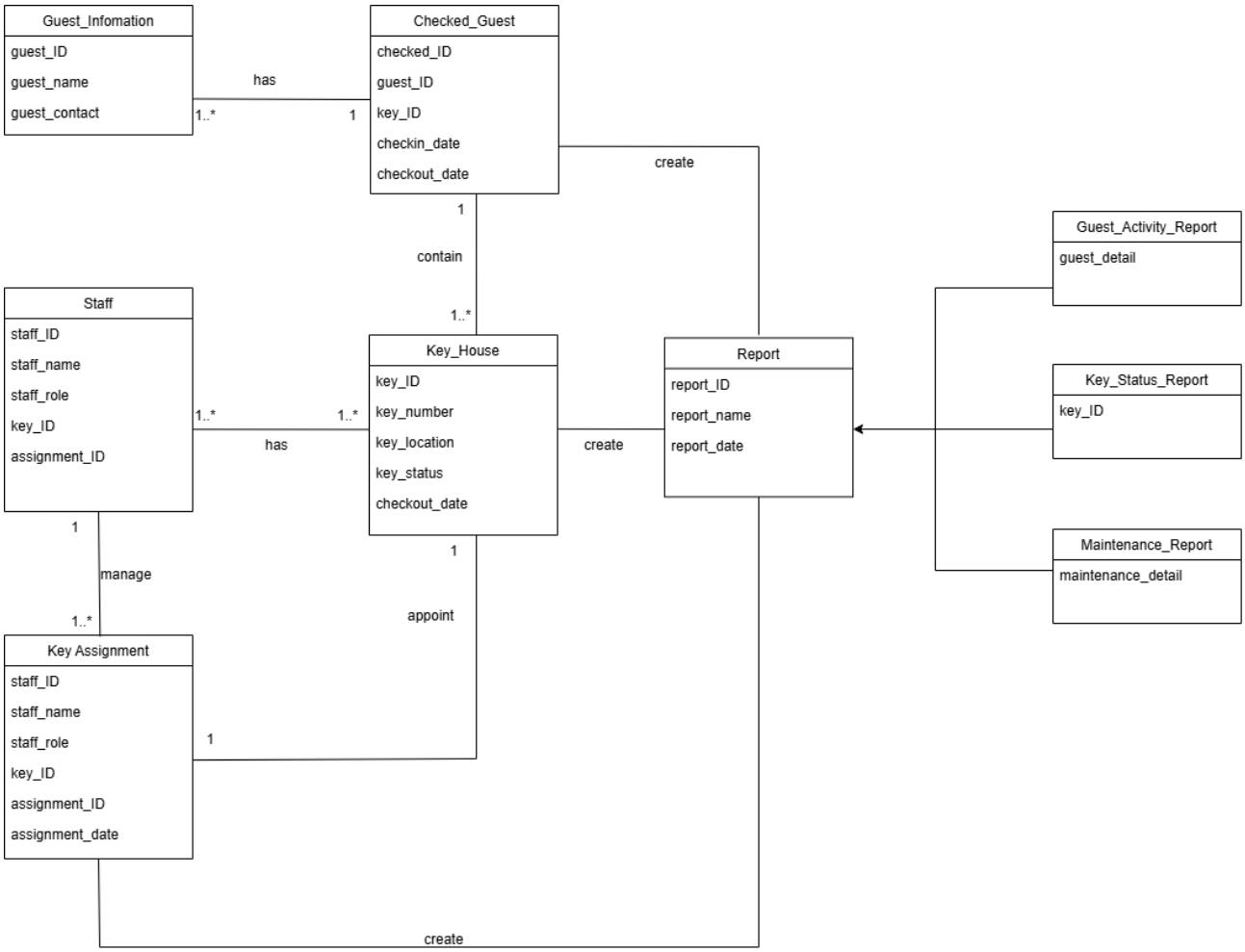
Business Activity Diagram: View Report



2. CLASS DIAGRAM

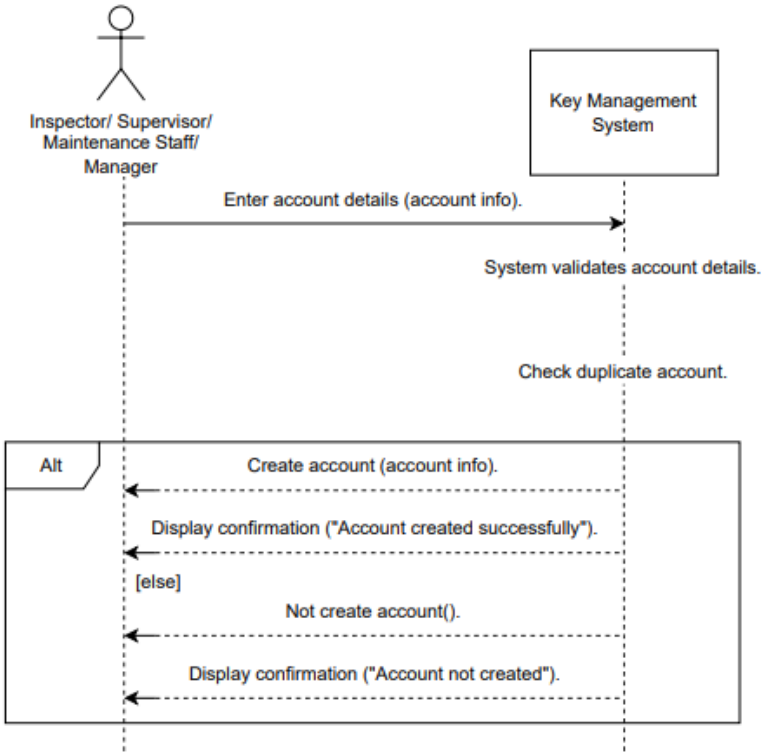
Domain Class Diagram

KEY MANAGEMENT SYSTEM

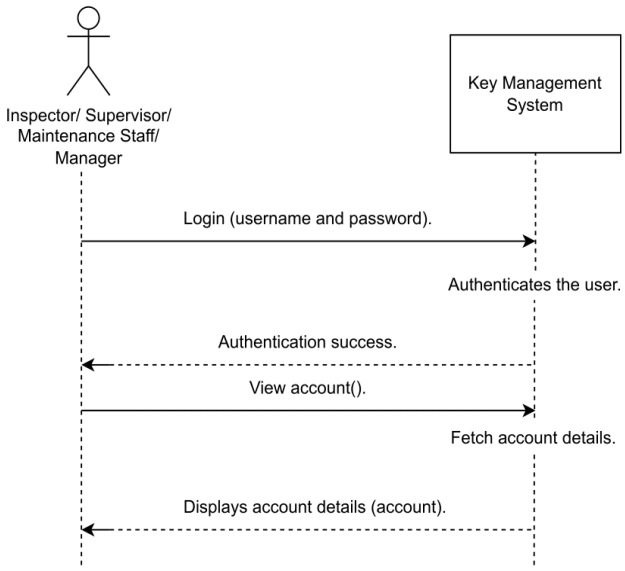


3. SYSTEM SEQUENCE DIAGRAM

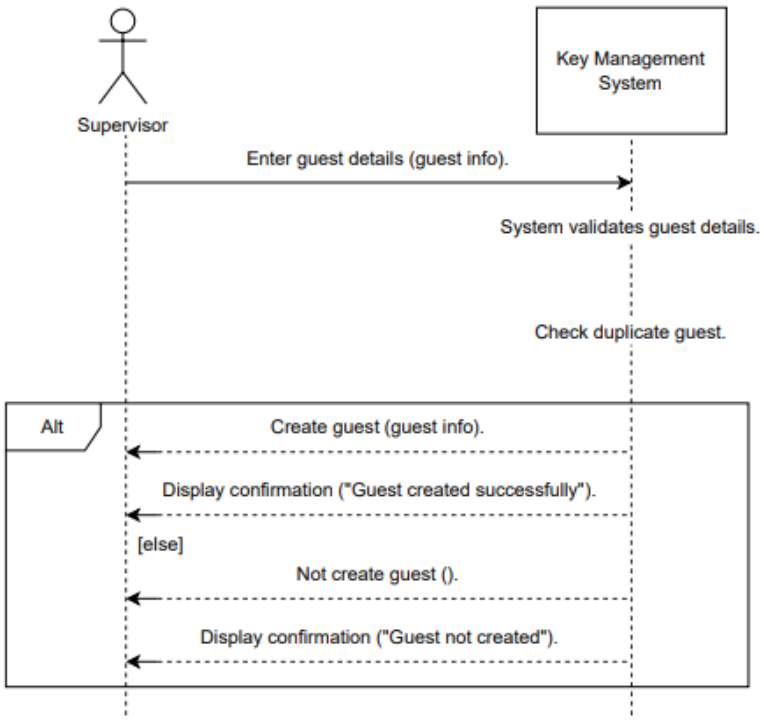
System Sequence Diagram: Create Account



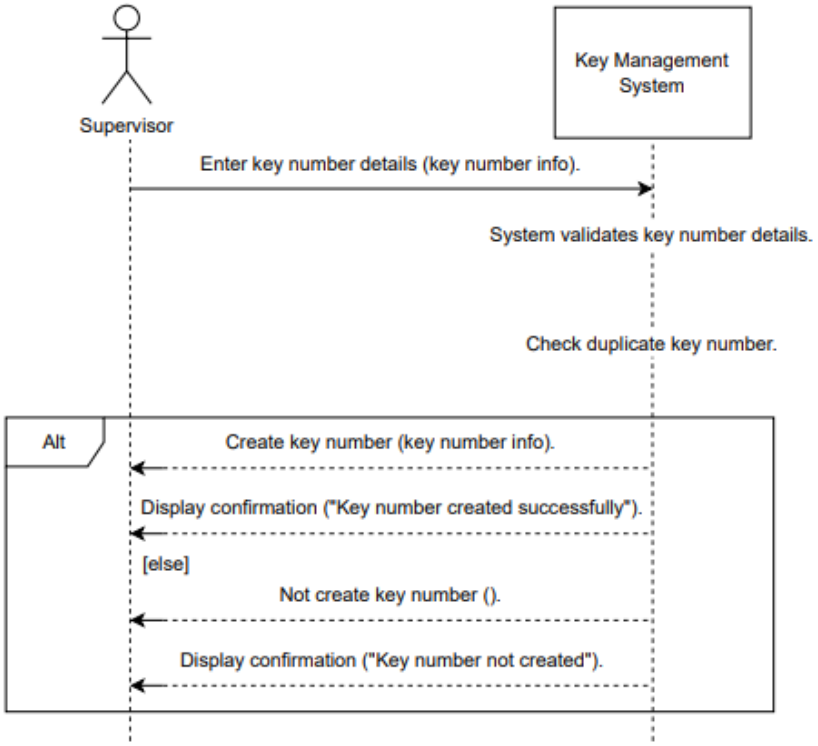
System Sequence Diagram: View Account



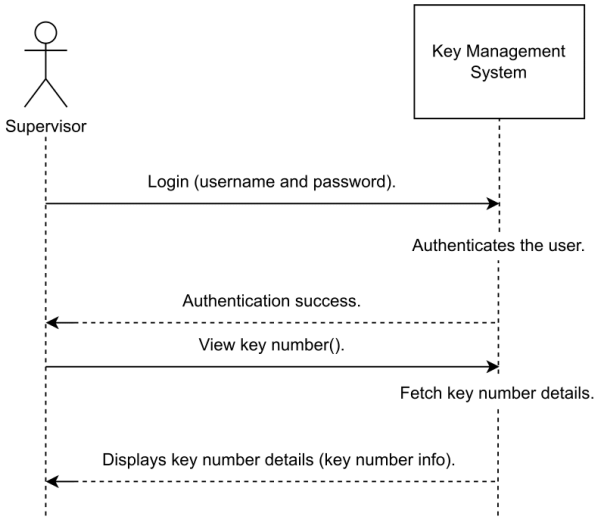
System Sequence Diagram: Create Guest



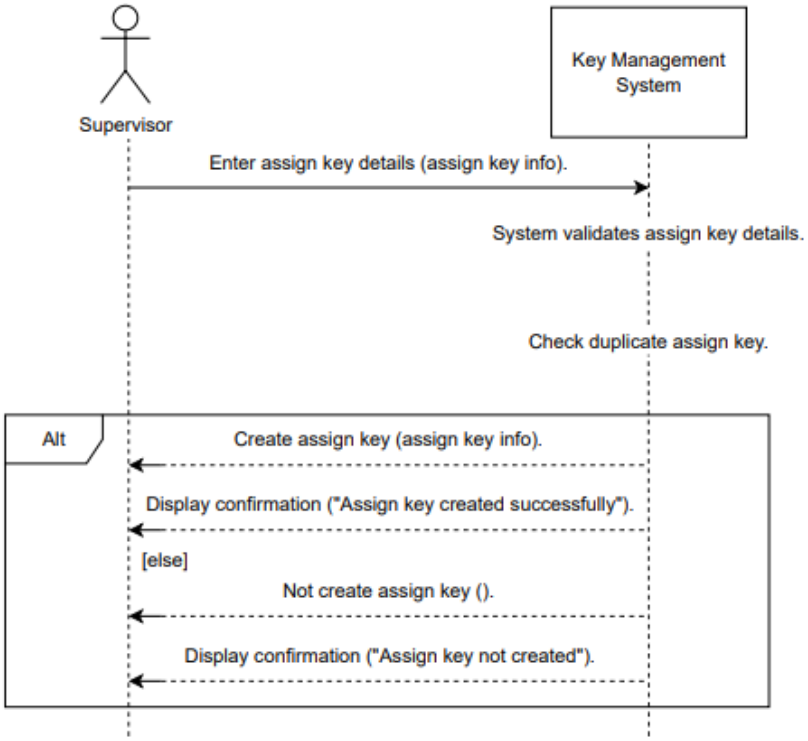
System Sequence Diagram: Create Key Number



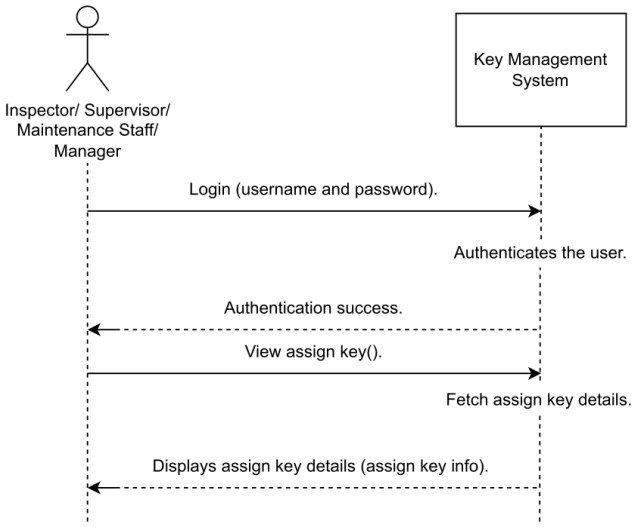
System Sequence Diagram: View Key Number



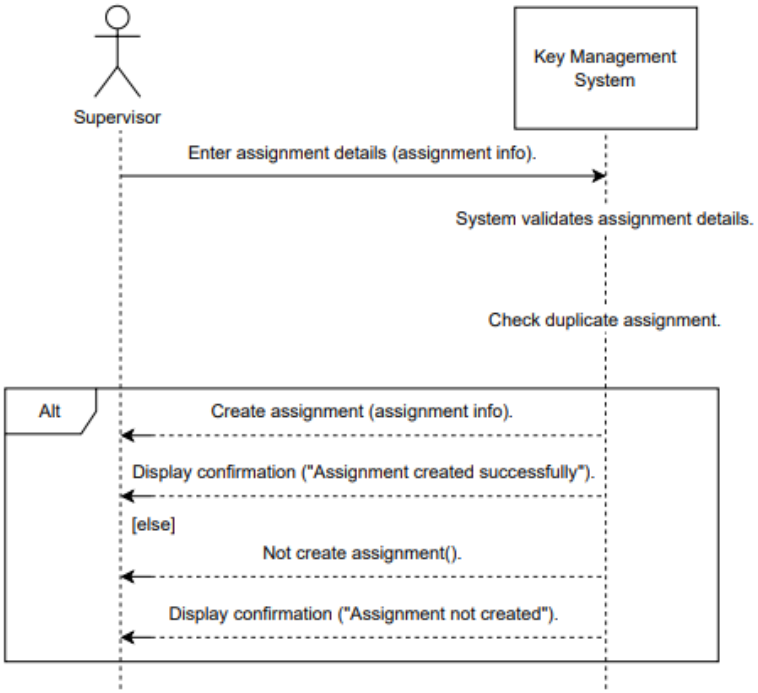
System Sequence Diagram: Create Assign Key



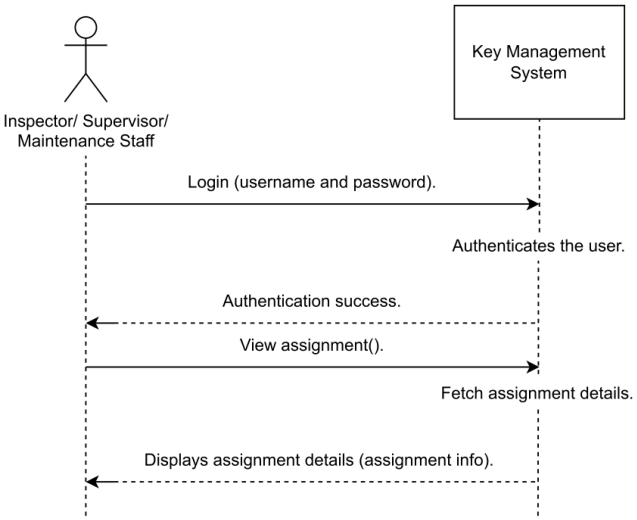
System Sequence Diagram: View Assign Key



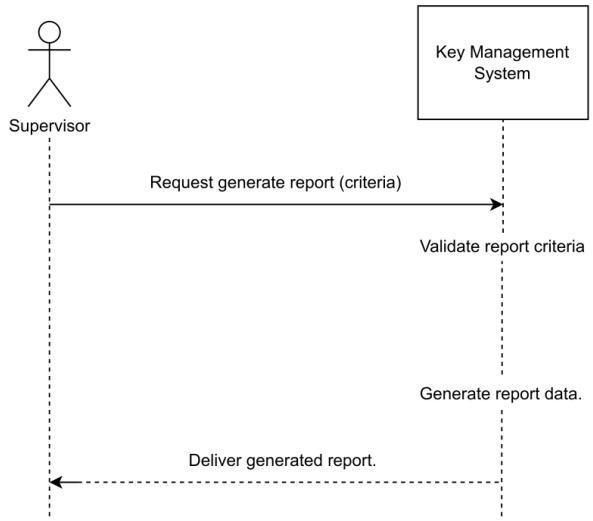
System Sequence Diagram: Create Assignment



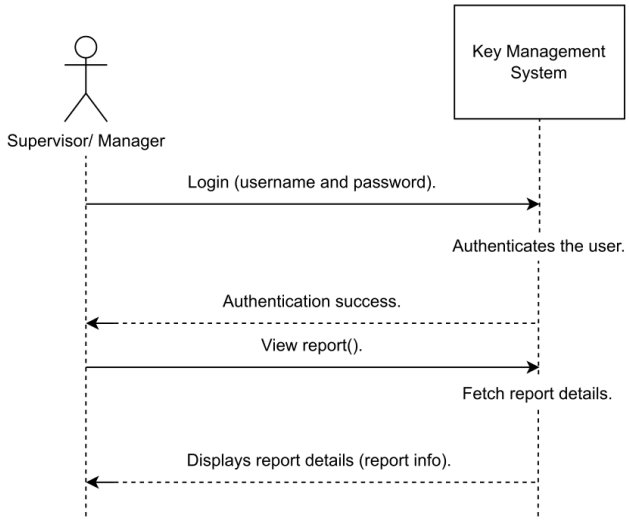
System Sequence Diagram: View Assignment



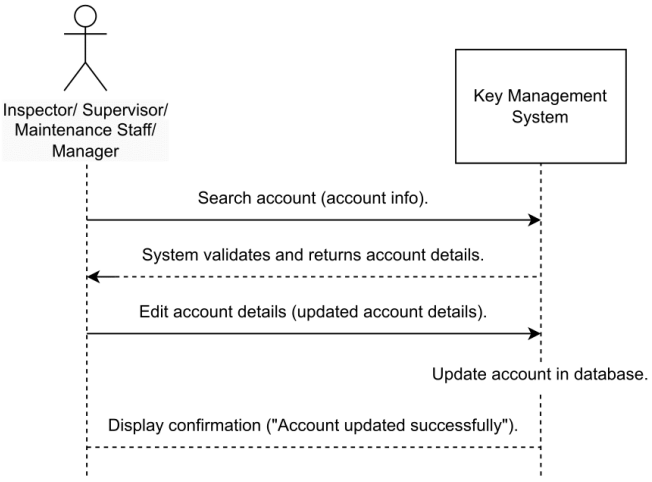
System Sequence Diagram: Generate Report



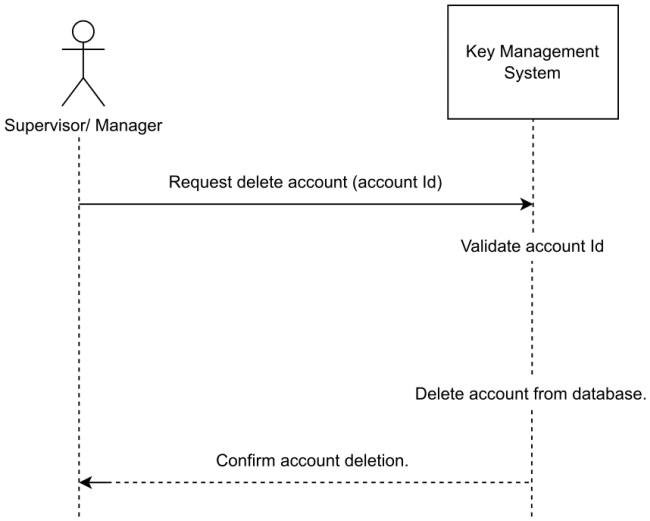
System Sequence Diagram: View Report



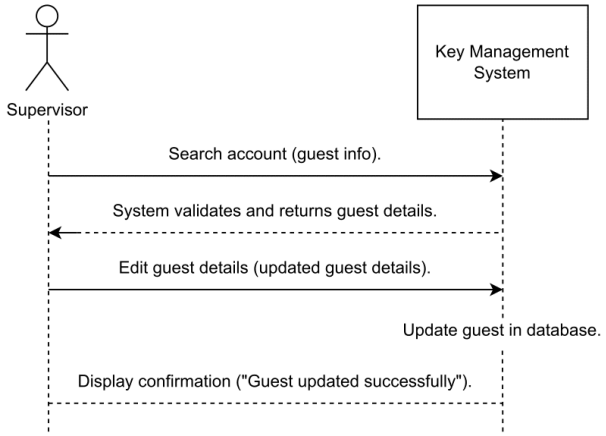
System Sequence Diagram: Update Account



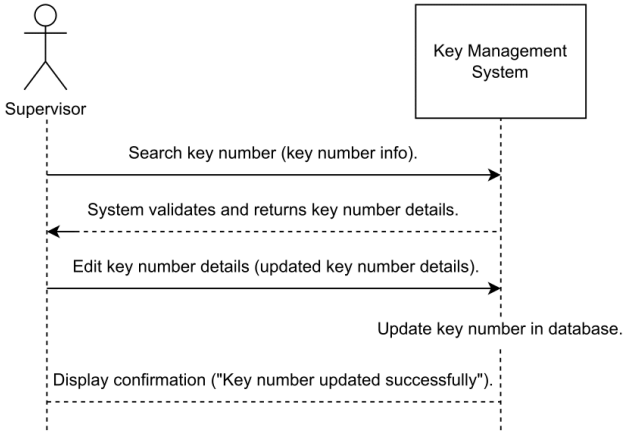
System Sequence Diagram: Delete Account



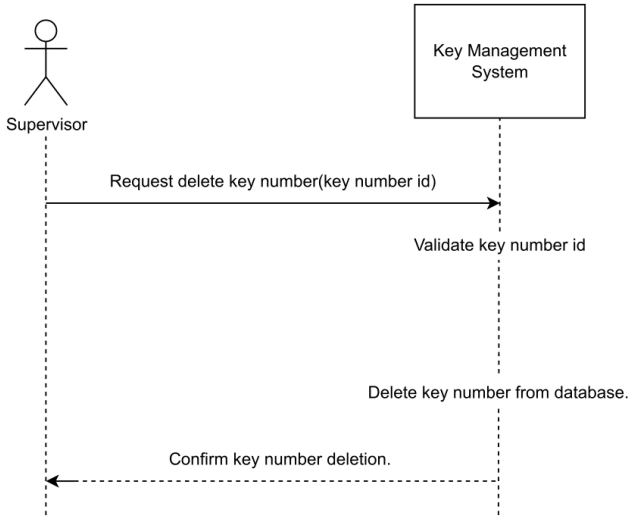
System Sequence Diagram: Update Guest



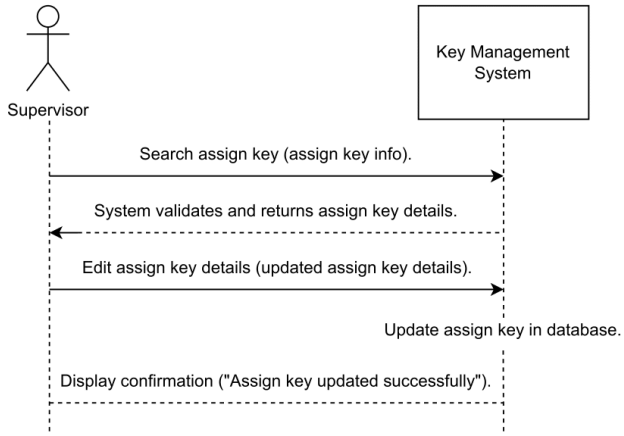
System Sequence Diagram: Update Key Number



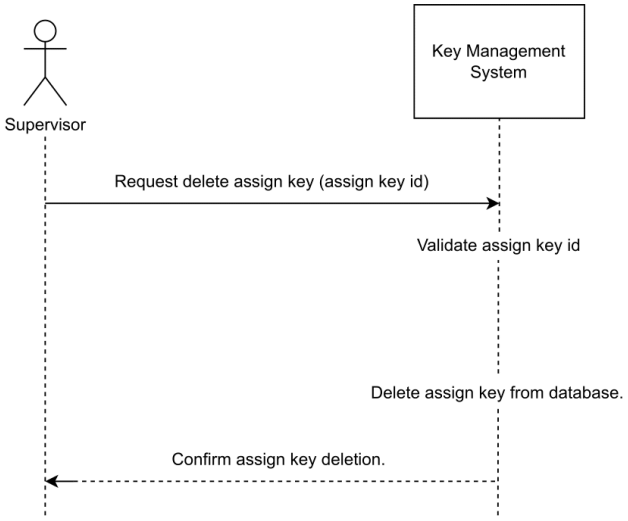
System Sequence Diagram: Delete Key Number



System Sequence Diagram: Update Assign Key

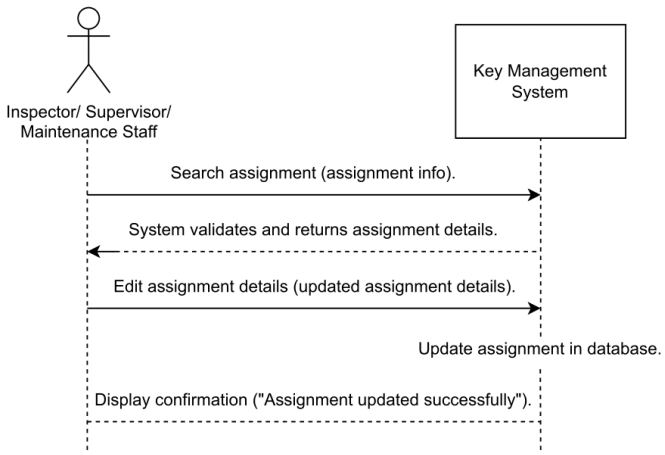


System Sequence Diagram: Delete Assign Key



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System Sequence Diagram: Update Assignment



Appendix B – Group Log

Group Meeting Log

Num	Date	Time	Details	Location
1	19/11/2024	11:00 AM	Interview with Stakeholder	Luzzeri Group, Kampong Bharu
2	21/11/2024	4:04 PM	Project discussion	Online
3	10/12/2024	9:30 PM	Use Case Discussion	Kolej Tun Sri Lanang A
4	17/12/2024	10:46 PM	Use Case Update	Kampung Mendapat
5	18/12/2024	12:11 AM	Domain Class Diagram Discussion	Kampung Mendapat
6	7/1/2025	12:48 PM	Use Case Description Discussion	Kolej Tun Sri Lanang A
7	7/1/2025	10:20 PM	System Sequence Diagram	Kolej Tun Sri Lanang A
8	8/1/2025	8:30 AM	Prototype Discussion	Bilik 24 Jam